

Drought Prediction using Deep Learning Framework with Multi-sensor Satellite Images

*Project Report submitted
in
partial fulfillment of requirement for the award of degree of*

**Bachelor of Technology
in
Computer Science and Engineering
by**

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May 2025

Department of Computer Science and Engineering

G H Raison College of Engineering

An Empowered Autonomous Institute affiliated to Rashtrasant Tukadoji Maharaj Nagpur University, Nagpur

Accredited by NAAC with "A++" Grade (3rd Cycle)

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Declaration

We, hereby declare that the project report titled “**Drought Prediction using Deep Learning Framework with Multi-sensor Satellite Images**” submitted herein has been carried out by us towards partial fulfillment of requirement for the award of Degree of Bachelor of Technology in Computer Science and Engineering. The work is original and has not been submitted earlier as a whole or in part for the award of any degree / diploma at this or any other Institution / University.

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Certificate

The project report entitled as “**Drought Prediction using Deep Learning Framework with Multi-sensor Satellite Images**” submitted by **Dev Sharma, Harsh Patel, Aayushi Dhote** for the award of Degree of Bachelor of Technology in Computer Science and Engineering has been carried out under my supervision. The work is comprehensive, complete and fit for evaluation.

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ABSTRACT

This project focuses on the prediction of drought conditions using a deep learning approach, leveraging spatio-temporal data from multiple satellite sources. A Spatio-Temporal Graph Neural Network (STGNN) is employed as the core model due to its ability to effectively capture both spatial and temporal dependencies within the data. Alongside the STGNN model a Long Short-Term Memory (LSTM) model was implemented to compare them, which gave predictions by processing sequential time-series data.

The primary goal is to improve the accuracy and timeliness of drought forecasts by analyzing soil moisture patterns. A hybrid dataset is constructed by combining observations from satellites like Sentinel-1, HLS, MODIS. Sentinel-1 provides radar-based SAR data, which is resilient to weather and lighting conditions and useful for estimating surface moisture levels whereas HLS offers a high-resolution optical data with vegetation indices like NDVI and EVI, which correlate with vegetation water stress, similarly we are using MODIS for using LST data which is crucial for monitoring heat stress in vegetation and the dryness of soils.

The data acquisition is implemented in Google Earth Engine (GEE) for efficient satellite data extraction and preprocessing, while Google Colab is used as the deep learning environment for training and evaluation. By fusing temporal sequences of satellite observations with spatial relationships between geographical regions, the STGNN model predicts drought levels or soil moisture anomalies across time while we also trained a Long Short-Term Memory (LSTM) deep learning model which was used to analyze sequential time-series data. This hybrid model helps capture complex interactions in the landscape, offering better generalization and insight compared to traditional models.

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CHAPTER1.
INTRODUCTION

1.0 INTRODUCTION

Drought is a consistent and devastating natural occurrence of the environment affecting food security and water access as well as the general ecological system. Excessive absence of rainfall is the primary factor for drought that can ruin crop yields and local supplies of water and put excessive pressure on rain-fed communities. On a global level on the upper hand drought is still an acute problem for which poor countries like India suffer most, as it's agriculture that underwrites the livelihood of a huge portion of its population. India's green thumb nature of relying on monsoon rains increases the impact of climatic changes on it. The irregular rainfall trend, delayed onset of monsoon, and protraction of dry spells have made many constituencies drought enclaves, with adverse consequences on agricultural output and sustainable agrarian economic equity. Notably, Maharashtra in general and its Marathwada region have become highly affected by drought and are even worsted by a number of other states. Marathwada region of Maharashtra is visualized in 'Fig. 1.1'.

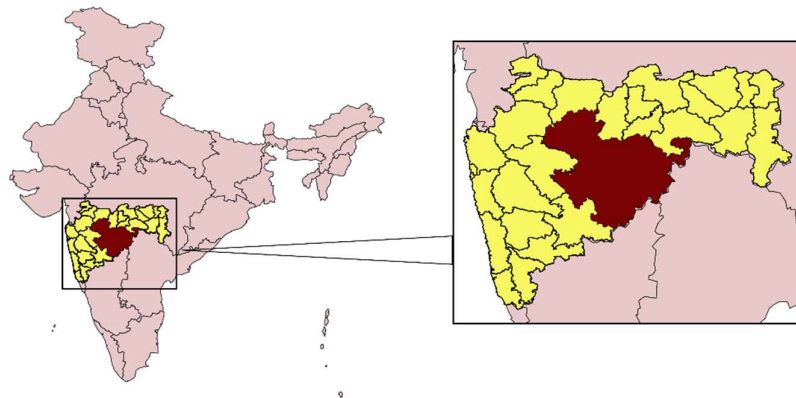


Fig. 1.1: Study Area

The Marathwada region has had a lot of problem arising from climatic unpredictability over the years. The semi-arid state of the area coupled with occasional and rogatory rain has become a restraining factor in the recurrence of drought. The region faced intense

droughts in 1972, 2000, and 2003, but the period from Such climate related catastrophes have weakened both agricultural output as well as caused enormous socio-economic hardships such as levels of farmer suicide, acute water scarcity and mass migration. Marathwada's rain-fed agriculture leaves its farmers in a position of considerable vulnerability, since farmers frequently struggle to acquire advanced technologies of irrigation, or reliable targets of weather patterns. With this, accurate and timely drought forecasting in Marathwada is found critically important for both scientific study and socio economic well being of people, to make decisions, minimize losses and make people more resilient.

Many ground-based and meteorology data-based drought evaluation methods are compromised by the small area of coverage and the delayed feedback. Old-fashioned manual monitoring is not time efficient and tends to find it difficult to maintain updated, comprehensive assessments of environmental conditions over large areas. The use of Satellite remote sensing has become a cornerstone contribution in drought monitoring, with the capability of ongoing and comprehensive tracking of environmental patterns along extensive geographic areas. Up-to-date satellite platforms carrying optical, radar, and thermal sensors allow for the diversification of data associated with critical drought indicators such as vegetation vitality, preservation of soil moisture, and changes of surface temperature.

Using advanced remote sensing devices and deep learning methods, this research attempts to improve drought prediction in Marathwada. For example, it uses data from the Harmonized Landsat and Sentinel (HLS), Sentinel-1 and MODIS satellite missions. Sentinel-1 provides Synthetic Aperture Radar (SAR) imagery, a unique asset with which to analyze surface moisture that is resistant to obstructions from clouds or change in the lighting conditions. HLS also provides high-quality data of optical for further derivation of indices that directly reflect plant health and water related stress, such as NDVI and EVI. MODIS also delivers thermal data and particularly Land Surface Temperature (LST) that is important indicator to measure heat stress and severity of drought.

To Applying advanced deep learning methods, obtained during the course of the study, including the Spatio-Temporal Graph Neural Network (STGNN) and Long Short-Term Memory (LSTM) networks, it is used to It is for this reason that the STGNN model excels here because of its ability to analyze both the shared connections bet we areas as well as the time patterns that unfold. This integrated approach improves its accuracy in modeling the propagation of drought relative to standard methods. STGNN offers a firm approach to drought trends detection and future development estimate by using graph structures for representing spatial relationship and integrating time-series data. At the same time, LSTM networks are used because of their expertise with sequential data. Because LSTMs are a subtype of RNNs, it is appropriate for long-term dependency sequences, enabling them to learn about the pattern of past climate trends and predict the occurrence of future droughts using the pattern of rainfalls and soil moisture index and vegetation index.

The results of this research help to address the critical need for strong, real-time, and geospatially detailed monitoring of drought in places such as Marathwada. The research uses satellite remote sensing and advanced deep learning methods to create a scalable and efficient drought forecasting mechanism. Exploitation of the merging STGNN and LSTM models, researchers develop a robust system that can identify regional connections and time-dependent trends that can guide decision making on agriculture, water use, and emergency preparedness. By using this strategy, scientists get more understanding of drought patterns and at the same time allow policymakers, farmers and communities to make informed decisions on safeguarding long term adaptability to climate change.

CHAPTER 2. LITERATURE REVIEW

2.0 LITERATURE REVIEW

Finding and solving drought has become a major area of research due to its tremendous impacts in society and environment. The application of satellite-based observations through remote sensing technologies has revolutionary transformed the ability to monitor droughts in time and space. The combination of radar and multispectral satellite platforms (the platforms that inspired such ones as Sentinel- and the Harmonized Landsat-Sentinel (HLS)[1]) is a significant breakthrough in the field of drought detection. These tools offer detailed, periodic and up-to-date information, and allow the precise measurement and identification of drought-affected areas using sophisticated analysis methods. A major investigation examined the possibility of utilizing Sentinel- SAR data to identify the damage caused by drought in temperate deciduous forests. Scientists studied differences in backscatter intensity from Sentinel – compared to optical vegetation metrics like NDVI and EVI. Analysis of data indicated that reduction in SAR signal intensity, particularly in VH polarization, during a drought is an indicator of reduced canopy moisture and biomass [3].

Using time-series SAR data, the researchers were able to track changes in forest stands and quantify drought severity in conditions of overcast skies that muddle optical sensor abilities. This approach uncovered important information about the deterioration of vegetation structure and enhanced the precision of forest health assessments during droughts. In conditions where continuous clouds and large areas of forest prevail, this methodology provides clear benefits over the classical optical methods.

A significant improvement to drought detection across various landscapes was attained through research that combined HLS and UAV imagery. In order to develop a robust series of vegetation indexes, the research employed the temporal data of Sentinel-2 at high frequency and the spatial accuracy of Landsat-8 in the HLS context. Satellite index anomalies were confirmed by UAV photographs that also showed close-up variation in vegetation stress. Using this dual-scale data fusion technique enhanced the precision of drought mapping, especially in agriculturally heterogeneous areas, where irrigation and

land use practices may differ. With an aid of UAVs for ground truth validation and high-resolution satellite data, the research highlighted the possibility to improve the real-time drought assessment systems, a practice that is especially useful for precision agriculture and early warning programs [4].

In order to develop crop-specific analysis using satellite datasets, a research project in northwestern Poland utilized both Sentinel-1 and Sentinel-2 imagery to monitor the winter wheat in drought conditions. The analysis of SAR backscatter intensity in combination with NDVI enabled the study to track the phenological cycle of winter wheat during the season. The analysis pointed out that SAR VH backscatter was very closely related to the changes in NDVI during critical drought periods, specifically during the flowering and grain-filling periods. Such observations enabled evaluation of the effect of drought on the crop yields. This study illustrated how easily accessible satellite data sets and limited field information can be used to monitor agricultural drought on an operational basis. The combination of radar and optical sensors allowed for the creation of a strong process for tracking crop development and determining stress levels in real time [5].

SAR data has been increasingly applied with deep learning techniques to improve hydrological drought analysis, particularly in water body conditions studies. Research focusing on the Amazon drought of 2023 used Sentinel- imagery and a U-Net segmentation model to monitor surface water variations within the Rio Negro basin. The study showed how SAR imagery is independent of daylight and clouds and thus provides unmatched potential for real-time water detection. Using a U-Net model trained with data from both before and after a drought, researchers were able to make accurate assessment and quantification of decline in river level. This approach helped to make timely assessment of the impact of drought on water resources and promises to work in evaluating floods and wetlands. Machine learning with satellite radar data provides a viable solution to hydro-logical risk monitoring and emergency response initiatives [3].

The research also revealed how European forests responded to drought over the years by examining temporal changes in SAR backscatter for various forest types. Drought impacts were estimated over several years with the help of observing changes in radar backscatter and comparing the data to meteorological drought indexes (standardized precipitation index SPI). Serious changes of SAR signals, particularly those recorded in coniferous forests, indicated the presence of canopy stress and loss of biomass during extended droughts. Through analysis of trends over time, this work was able to distinguish between transient and long-term stress, thereby offering useful insights for long-term ecological studies. These findings are of great importance to forest managers and policy makers as they work to conserve biodiversity and maintain ecosystem robustness in the midst of increasing drought conditions [3].

Through the integration of high-resolution satellite data with climate indicators, a lot of advances have been made in the evaluation and forecasting of drought risks. An investigation introduced a hybrid modeling system that combined Sentinel-2 derived NDVI and NDWI indices and SPI data for assessing spatiotemporal drought risk in semi-arid zones. By combining harmonized data on vegetation and water, the approach was more capable of tracking anomalies that signal moisture loss and plant stress. By integrating the meteorological measurements, the study generated a complete drought map that was able to distinguish agricultural from hydrological drought events. Its wide applicability and need for no on-site data show the usefulness of the tool in the operation of early warning systems in transboundary drought identification. Although SAR was not directly used in the study, the developed approach can be easily combined with HLS and Sentinel imagery for wider usage [7].

Agricultural investigations identified evapotranspiration (ET) that was monitored using Sentinel-2 imagery and land surface temperature modeling, as a critical measure of drought. Satellite-derived ET anomalies were excellent indicators of crop water stress, giving an early nutritional warning of drought before visual changes became apparent. In order to support irrigation management, the team developed a decision-support

approach based on analysis of spatiotemporal ET patterns. Although in-situ adjustment was required, the approach showed great promise for improving precision farming and for improving the utilization of water resources. The research clearly indicates that the integration of thermal-optical remote sensing with the structural information from Sentinel- SAR can improve monitoring of the initial stages of drought[6].

There has been a significant growth in drought classification through the application of machine learning methods, partly because of their capacity to process diverse multi-sensor data. One notable success involved the use of a Random Forest classifier in aggregating Sentinel-, Sentinel-2, and Landsat-derived indices to delineate drought prone areas in southern India. Model validation against MODIS drought products and field samples showed the model accuracy above 85%. Apart from demonstrating that open-access satellite data can be applied to drought mapping, the study also proposed a reliable approach of integrating radar and optical variables. By combining Sentinel-VH/VV ratios and Sentinel-2 red-edge bands, the study precisely distinguished between drought-related change from other plant problems including disease or harvest[4].

Using Sentinel satellite data, researchers now study the manner in which urban green infrastructure manages heat and moisture stress events. By employing Sentinel-2 time-series data, a southern Spanish study measured trends in urban parks and green corridors. Researchers used NDVI profiles to determine how the health of green infrastructure in urban setup was conserved during long periods of drought. Although the primary focus of this paper was the urban planning, it focuses on how Sentinel data can be used for microclimate examination, which may contribute to the development of drought adaptation measures that are targeted. Integrating Sentinel data, this study extends drought analysis from agricultural and forestry sectors towards urban ecosystems[2].

Drought early warning was enhanced by a study that combined satellite indices and socioeconomic vulnerability factors. Based on a combination of Sentinel- and 2 satellite

data and data on crop insurance, irrigation, and population density, the authors created a complete drought vulnerability map. By using this comprehensive risk assessment on Ethiopian regions, researchers have shown the role remote sensing plays in improving disaster preparedness. It went further to address a major criticism of works that are circumscribed to satellite images by including human and economic vulnerability indicators in drought analysis. The merger of physical and social data, although technically complicated, marks an important transition from the existing drought risk evaluation approaches.

Studies show that although vegetation indices such as NDVI are common in drought detection, they are not effective when used separately. Several drought events in South Asia were exploited by the researchers in order to evaluate various indices like NDWI, VCI, and TCI. Indices were computed by both Sentinel-2 and Landsat-8 data sets, and their accuracy was subsequently confirmed by meteorological records. Research showed that NDVI always lagged behind so as to detect early signals of drought, especially during times when some vegetation was still active. Research suggested using combined index models together with early Sentinel-derived SAR data for detecting structural damage, a method that can be applied to your project's Sentinel and HLS data use [2].

Sentinel-1 SAR data were also used to detect drought-induced land degradation in arid and semi-arid regions using a novel approach. Researchers used an object-based scheme that combined SAR backscatter intensity data with texture features such as homogeneity and entropy to monitor soil moisture and vegetation cover change. The authors used both VH and VV polarization data to distinguish between thriving and stressed vegetation thus offering a reliable method of observing land degradation during droughts. When backed by HLS data for comparative ground truth reference, the method provides a potent means of monitoring desertification at a large scale within drought-prone regions. The multi-layered classification systems are very promising to the implementation of operational drought monitoring programs[1].

Scholars paid attention to ecological regions to explore the potential of Sentinel-1 and Sentinel-2 data in detecting the impacts of droughts in a wetland ecosystem. The investigation used backscatter signal analysis and vegetation indices to track the amount and state of wetlands in the Amazon basin during a severe drought. The data revealed that wetland areas reduced significantly in size, accompanied by a reduction of biodiversity and ecosystem services. By incorporating radar and optical data, this study highlights the need of monitoring natural water bodies during droughts in order to understand water dynamics better. The results show that wetlands, which are critical in supporting biodiversity and carbon sequestration, need to be carefully monitored during drought events [5].

Another study in coastal situations measured the sensitivity of mangrove ecosystems to drought events with Sentinel-1 synthetic aperture radar (SAR) imagery. The observation was intended to monitor changes in mangrove canopy organization and the level of water scarcity associated with drought. By employing SAR backscatter variation, the study successfully identified locations where mangrove decline and canopy density reduction occurred. This work highlighted SAR's potential to observe subtle alterations to coastal ecosystems, and saltwater intrusion in drought exacerbates water stress further. Such an approach can be of great value for monitoring and protecting sensitive coastal ecosystems that are threatened by global warming and drought. Application of Sentinel-1 data to these ecosystems shows the applicability of the SAR in monitoring diverse habitats [9].

An important research improved the detection of droughts in urban areas through the use of Sentinel-1 radar data together with the Surface Urban Heat Island (SUHI) effect for monitoring drought stress in cities. The researchers used SAR-derived surface temperature anomaly maps to indicate hotspots in urban heat islands that were at high risk of water stress. The study showed that Sentinel-1 can help identify weaknesses in urban areas to drought situations by looking at the data with parameters such as green space and impervious surfaces. Incorporating Sentinel- science as part of urban

monitoring has introduced a radical approach for measuring drought vulnerability in cities as a basis for advanced adaptive planning initiatives [11].

New approaches in drought monitoring are emerging in the form of combined use of Sentinel- and hydrological models to measure drought conditions. Researchers used a hydrological model with information from Sentinel- SAR and global climate sources to estimate stream flow reductions from droughts in South Africa. By integrating Sentinel-2 SAR data with traditional climatic data, the study showed that models improve in their ability to simulate models of soil moisture levels, which are key to drought impact understanding. The incorporation of remote sensing technologies and predictive modeling offers significant benefits for water supply and environmental services estimates of how drought will impact river-based communities. Through remote sensing and hydrological models integration, researchers can better predict drought impacts, thus assisting policymakers' efforts to develop structured drought management strategies [10].

A study was conducted to improve the evaluation of drought-affected vegetation stress through understanding the seasonal greening and senescence timings through the use of Sentinel-2 and SAR satellite data. The investigation proved that monitoring changes in the stages of plant life by means of phenology provides an effective method of early detection of drought-related stress before the physical damage becomes visible. By combining optical and radar observations, the team realized that they could detect the early symptoms of vegetation stress better than using conventional NDVI approaches. The results highlighted the fact that the use of SAR with optical data greatly enhances the capacity of early warning systems especially for areas that experience consistent or changing drought situations[11].

Several studies have pointed out that machine learning has been used in combination with Sentinel- and Sentinel-2 data for the prediction of drought impacts. Researchers in western Australia used SVM (support vector machines) to determine drought affected

areas combining the Sentinel-1 and Sentinel-2 SAR backscatter and optical data. The SVM algorithm with the help of elements such as vegetation indices, soil moisture and surface roughness was able to predict the spread and severity of the drought. This investigation demonstrates the flexibility of machine learning approaches to work with multi-source datasets, and implies that these methods are well-suited to scaling drought surveillance in complex, broad areas. Additionally, it displays the benefits of machine learning with radar and optical technology for measurement of drought impact across large geographical areas[9].

Using Sentinel-1 observations, scientists have shown how the SAR technology can successfully monitor the impact of drought on forest canopies in Mediterranean climates. Using Sentinel-1 radar data, the study showed that prolonged drought events led to substantial changes in canopy moisture and biomass loss through changes in radar backscatter. It turned out that VV polarization was particularly sensitive to changes in forest structure and enabled the differentiation of healthy and drought-stressed vegetation. This research supports the idea that SAR is becoming an important tool for monitoring large-scale vegetation dynamics, and that its use with other remote sensing systems leads to effective and accurate assessment of drought [2][3][4].

Besides observation of vegetation, radar data usage in hydrological systems drought measurement has evolved. An analysis was done to find out how Sentinel-1 SAR data can be used to measure the magnitude and severity of drought in Central Asian reservoir networks. Discoveries showed that radar backscatter can be used to identify reductions in reservoir and lake levels of water, and monitor the loss of soil moisture in the surrounding agricultural fields. In arid areas with water shortages, this technique outperforms conventional optical sensors, which are typically plagued by persistent cloud obstruction. This research emphasizes the utility of the SAR technology for the management of water resources, especially during periods of scarcity of water and drought [5].

A thorough research project examined the use of deep learning combined with satellite data to evaluate drought conditions from Sentinel-1 and Sentinel-2 data. The study used a convolutional neural network (CNN) to process temporal data from Sentinel-1 and Sentinel-2 satellites in order to allow automatic mapping of drought-affected areas in Brazil. By the means of the detailed spatial resolution of SAR from Sentinel-1 and the extensive spectral data of Sentinel-2 optical sensors, the CNN model was able to detect the early drought signals and Automating the detection phase and reducing human participation are what make this mechanism especially useful for real-time drought monitoring and response activities. The performance of convolutional neural networks (CNNs) for processing diverse remote sensing data demonstrates that artificial intelligence has become ever more important for environmental monitoring [8].

To explore the impact of drought on biodiversity, researchers used Sentinel-1 radar backscatter data to monitor forest ecosystem health in the Mediterranean Basin. The research showed the temporal acuity of SAR measurements in detecting early vegetation stress due to drought especially in areas with a mix of vegetation types. The study identified forest dieback using changes in radar backscatter coefficients, a finding that was supported by follow-up ground investigations and field data. The ability to detect small changes in canopy structure is key to conservation work, particularly in ecosystems where long duration drought may pose a risk. Such results demonstrate how Sentinel-1 data provide valuable insight on tracking ecological dynamics over large areas [7][8].

Sentinel-1 data, in combination with climate models, played a fundamental role in enhancing the prediction of severe drought in river basins. In India, investigations used Sentinel-1 data to monitor soil moisture on the Ganges River Basin and further applied this information for hydrological models to forecast streamflow. This investigation showed that Sentinel-1 imagery, in combination with climate models, provides far greater support for accurate climate predictions in regions with constrained ground observation. Application of this approach significantly enhanced timely warning systems and aided better water resource management decisions during droughts. The

combination of satellite radar (SAR) and climate predictions is used to determine viable strategies for fighting water scarcity in extreme droughts [4][5].

Scientists used Sentinel-1 and Sentinel-2 data in a pioneering study to determine the potential of radar and optical imagery in monitoring drought in East Africa's agro-pastoral regions. Under a multi-sensor approach, the study examined the status of vegetation and water availability. By correlating variations in SAR backscatter with optical vegetation indices variations, the study strengthened the drought monitoring system such that it is applicable in crop cultivation and livestock management. The effectiveness of such an approach illustrated the significance of using various types of data for a more comprehensive assessment of the drought impacts in different land use settings. In agricultural regions particularly vulnerable to climate change, the use of multi-sensor methods might be especially important [12][13].

Sentinel-1 data were also used to compare the impacts of drought on the Australian outback, focusing, in particular, on the relationship between SAR backscatter intensity and soil moisture. A time-series analysis of SAR imagery was used by the investigation to identify diminishing soil moisture areas and correlate them with real reports of drought conditions. The study indicated that both VV and VH polarization SAR backscatter was useful for understanding variation in soil moisture hence a good way of assessing water resources in arid locations. Because dryland areas are water scarce and have a need for reliable drought prediction systems for the welfare of the community, this work is of critical importance [11][14] .

Besides, scientists integrated Sentinel-1 and HLS data to monitor vegetation health changes due to drought in the Sahel region. Scientists combined SAR derived soil moisture information with HLS derived vegetation indices to detect the stress in the vegetation and predict the onset of drought. It was shown that the high temporal resolution of HLS and Sentinel's moisture sensing capabilities made the two platforms complementary for seasonal drought monitoring in arid regions. This strategy is very

beneficial to Africa as it is often overwhelmed with inadequate ground-data for drought monitoring [12].

By using the multi-temporal SAR data from Sentinel-1, another study examined the changes in the tropical forest canopy height due to drought. The study showed the potential of using analysis of changes in radar backscatter over time to estimate canopy heights, which is an essential indicator of drought stress in forests. The combination of field data with Sentinel-1 radar imagery showed that this system is very appropriate for monitoring the health of large-scale forests during droughts, particularly when areas are densely covered or there is limited access [13].

Further research used Sentinel-1 SAR data to study the impact of drought on water resources in the Colorado River Basin of the United States. Examination of the radar signal from the Sentinel-1 data aided the study in detecting declines in surface water at critical reservoir locations. This method acted as an early warning system for water shortage resulting from drought hence qualifying to deliver essential benefits to water management of extreme droughts. This experiment demonstrates how Sentinel-1's radar backscatter can be used to provide useful information about hydrological systems and improve the early detection of water scarcity.

Researchers were able to create a deep learning algorithm, using Sentinel-1 and Landsat data, to predict vegetation water stress in arid areas. The team trained a Convolutional neural network (CNN) model with large satellite data from both sources, and they successfully predicted vulnerable areas to severe drought conditions. The model was effective in determining locations with reduced water availability in vegetation coverage, mostly in drought-prone areas. Using this technique, researchers emphasize the potential of deep learning to work with diverse satellite data, which finally contributes to the creation of more accurate tools for drought assessment and monitoring [14].

Additionally, a thorough analysis used Sentinel-1 SAR data to determine how drought impacts wetlands in Southeast Asia. SAR backscatter data was overlaid with indices for flood and drought monitoring to determine changes in wetland ecosystems. The research showed that Sar's sensitivity to small differences in water level and vegetation status meant that it was very effective for wetland ecosystem surveillance. Consistent and precise monitoring of wetland health is essential in coastal regions with long-term droughts, as these occurrences may have serious ecological and socio-economic consequences [10].

>>Researchers recently assessed the integration of SAR and optical satellite data to enhance the effectiveness of early drought warning systems. Using Sentinel-1 and Sentinel-2 satellite data, the investigation proposed a holistic approach to monitoring drought at a high frequency, not only estimating the health of vegetation, but soil moisture as well. The analysis revealed that SAR data, because it can measure soil moisture variability, is in a position to complement the understanding of optical data that is more premised on the way vegetation is affected. The two data types were combined, producing better accuracy in drought condition detection especially in areas where cloud cover often hampers optical imagery. Such an integrated sensor approach could make global drought monitoring activities immediately updated and better prepared for the upcoming drought phenomena.

Major developments have arisen from research investigating the use of machine learning algorithms to predict vegetation water stress using data from several remote sensing sources. Using random forests and support vector machines (SVM), scientists categorized dratted zones in Central Asia depending on Sentinel-1 SAR and Landsat-8 datasets. Using radar backscatter shifts and optical vegetation indices, the model was able to identify areas suffering from severe drought stress. The results indicated that the combined use of SAR and optical data produced better classification accuracy than using a single source of data. This shows that combining different remote sensing measurements and using

machine learning are critical to improving drought monitoring and enhancing early warning capabilities.

Fluctuations in soil moisture resulting from drought are frequently hard to detect, especially in forested or hilly regions. Using radar capabilities of Sentinel-1, scientists were able to visualize changes in soil moisture in forested and mountainous regions where optical monitoring could fail. The research used time-series analysis to track changes in soil moisture and detect vegetation water stress in drought-affected zones. By exploiting its radar backscatter sensitivity to moisture, Sentinel-1 allowed for the detection of minute changes in soil moisture, which are critical measurements of drought effects. The results show the applicability of radar remote sensing for environmental monitoring in areas where optical data is limited by such aspects as cloud cover or dense vegetation [12].

In the tropical areas, the application of Sentinel-1 SAR data was extended to investigate drought-induced pressures on the forest ecosystems. Radar backscatter data acquired in Amazon forests showed a decline in canopy moisture and biomass during the drought period. The researchers used temporal patterns in the radar data to monitor shifts in the vegetation health, highlighting the usefulness of SAR remote sensing in determining the impact of drought on forest structure. Remote monitoring of forest health is critical to successful sustainable forest management, especially in tropical regions where drought may have drastic ecological and economic consequences.

By combining Sentinel-1 data with crop yield forecasting models, a major milestone was achieved to monitor the impacts of drought on agriculture. A study of agricultural drought detection in the Midwest United States used Sentinel-1 SAR data to derive estimates of soil moisture and crop health during a period of drought. Satellite-based estimates of soil moisture derived from SAR were shown to correlate with reductions in crop yields, demonstrating that Sentinel-1 has the potential to facilitate precision agriculture during periods of drought. This study represents the potential for near real-

time food security monitoring and the monitoring of crop yields using satellite-derived data to improve on food security management and seasonal agricultural planning during periods of drought.

In the field of hydrology, Sentinel-1 data provided the basis for a study in which researchers monitored the environmental impacts of drought on river systems, where the fluctuations in surface water were the main interest. Through the examination of backscatter data over a year of drought, the researchers were able to monitor changes in surface water levels over multiple rivers. The study demonstrated that Sentinel-1 SAR can monitor the depletion of water from rivers, which proves to be important for environmental drought monitoring. The results from the work can improve on the understanding of flood forecasting and drought prediction modelling particularly in areas where river discharge is prominent in understanding hydrological stress and water availability [17] .

Another paper that is relevant evaluated the capability of Sentinel-1 to monitor drought in the context of water stress in big cities. The researchers studied the capability to harness SAR data to evaluate the impact of drought on urban water supplies through the assessment of variation in surface water and soil moisture levels in the city of Cape Town, South Africa. The study shows that the SAR data could effectively monitor the decreases in surface water and the changes in soil moisture. Thus, the SAR data could efficiently monitor the urban drought. As urban areas face increasing water shortages, this study will provide an important facility to manage their drought risk.

Research has documented the use of vegetation phenology for drought monitoring. The integration of Sentinel- 1 SAR data with vegetation phenology models aims to examine the interrelationship between vegetation health and drought severity. By using the model for SAR data in the Mediterranean, one can monitor phenological events including the timings of leaf expansion and senescence, as stated by the researchers. The first sign

of drought stress in the vegetation is the phenology change. When used along with radar data, the application of phenology alteration can enhance early drought warning.

Regarding methodological developments, A study investigating the Sahel worked with the potential for integrating HLS data with Sentinel-1 radar backscatter to develop a combined drought monitoring system. The project team synthesized time series data from both sensors to map vegetation stress and water deficits. Overall, the study found that frequent high-quality data offered by HLS were necessary for monitoring vegetation health, while Sentinel-1 radar data clarified soil moisture and surface water availability. The combined use of the two sensors was especially useful in the Sahel where seasonal drought expressed a major risk to agricultural productivity and food security.

One contribution of note investigated the utility of satellite remote sensing data to enhance socio-economic vulnerability indices for improved drought risk management. The authors used Sentinel-1 and HLS data to determine areas of agricultural drought susceptibility in southern Africa while accounting for features like population density and poverty maps to create vulnerability maps. The paper indicated that satellite remote sensing data and socio-economic indicators actually worked together in examining the broader impacts of droughts on vulnerable populations. A multi-faceted approach encapsulated drought with risk provided a better view of drought risk and would be a valuable tool for developing policies and interventions as needed [1].

A study that worked in the Mediterranean region, based on Sentinel- 1 SAR data combined with drought indices to track long-term drought conditions. That study used some of the differences in radar backscatter value over time to look at the connections between soil moisture, vegetation health, and drought severity. The study concluded that the Sentinel- 1 data noted decreased vegetation health parameters well before conventional drought indices would reflect in terms of declaring an agricultural drought. As a result, Sentinel- 1 data could serve as a tool for more timely, proactive drought management in semi-arid areas. This study underscores the increasing utility of radar-

based remote sensing to monitor and manage drought risk in climate-sensitive landscapes [12].

As remote sensing technology develops, a study examined the potential to leverage the SAR data of Sentinel-1, with the high-resolution optical data from Landsat-8, for monitoring drought impacts in agriculture agglomerations in California. Using several images over distinct periods, the study could establish drought stress indicators for crops through differences in vegetation index and soil moisture content. The findings indicated that the combination of SAR data with optical data was very useful in establishing droughts, because SAR can locate changes in soil moisture content and optical data can provide details of vegetation health. Thus, this method has potential for wide-area high-resolution drought monitoring, and can also have a role in crop yield forecasts and assessment for water resource management.

An important advancement in drought monitoring was a research study in the Amazon rainforest looking at Sentinel-SAR data through drought effects on biodiversity and forest structure. The study of radar backscatter of a forested area during multiple times of an extreme drought showed canopy moisture deficits and changes in biomass, both reflecting drought eco-physiological stress in tropical forest. The authors stated that they had shown SAR, specifically the sensitivity to a vegetation's water content, could help monitor forest health during drought. The authors ultimately concluded they observed important findings on the need for continual observations of forest to understand (1) whether prolonged droughts produce ecological effects in the tropics, especially within a context of global warming.

Following of study, a separate study investigated the utility of SAR data to monitor soil moisture decreases in a drought in semi-arid regions. This research used Sentinel-1 to monitor moisture loss in Southern Spain soil moisture across soil covers (grasslands, croplands and forests). They were able to use data of Sentinel-1 SAR backscatter accurately to detect soil moisture deficits in areas of mixed vegetation in the semi-arid

areas. This study added more evidence formative of SAR as useful in monitoring soil moisture deficits during drought, especially useful for monitoring due to the international importance of knowing water availability for agriculture, natural vegetation and planning if use of land.

In a related study, drought detection in hydrological systems used Sentinel-1 SAR to monitor changes in water bodies and river discharge in central Europe. As part of their study, combining SAR and river discharge found changes in river volumes from prolonged drought. Results indicated that SAR data could detect subtle changes in water bodies, providing useful data for hydrological drought detection. This method can improve the predictive utility of drought on hydrological systems, which will help decision-making in managing water resources available during these critical periods of drought. [10]

In another study that expanded on the work of Sentinel-1 SAR, satellite data was used to estimate the magnitude of water loss in wetlands due to drought. This study conducted in Southeast Asia explored the changing water levels and vegetation coverage of wetland ecosystems during dry periods. Researchers concluded that as drought decreased available water it decreased wetland area due to drought-induced drought-induced water availability. Remote observation of these important environments is critical to assessing the future implications of drought on biodiversity and local communities, as these wetlands are often located in water stressed regions [14].

In terms of machine learning application in remote sensing for drought detection, a study used deep learning approaches based on machine learning with both Sentinel-1 and Sentinel-2 data to analyze and predict drought impact to vegetation cover in Central Asia. The authors trained a neural network to identify land cover affected by drought by using both temporal Sentinel-1 and Sentinel-2 data. The model was able to classify the presence of drought with high accuracy when using the both the optical and radar data in the same model. In a similar vein, this highlights the growing role artificial intelligence has in enhancing remote sensing for drought detection. Using these types of machine

learning algorithms, these approaches could drastically improve drought monitoring and prediction models, which would will also provide near real-time monitoring of drought conditions.

Further, a valuable study utilized Sentinel-1 and Sentinel-2 data to develop a multi-sensor drought monitoring framework for sub-Saharan Africa. Using radar and optical data, the authors were able to assess both soil moisture and vegetation health, thereby increasing spatial extent. By using multiple datasets, they successfully produced improved drought maps, while at the same time increasing spatial resolution of their drought monitoring. This example shows the immediate utility of integrating data from sensors with other metrics, indicating faults in ground data such as in regions where data exists or where ground data are poor or imprecise.

An analogous study in the Mediterranean area also concentrated on the combination of SAR and optical datasets to study how droughts could be monitored more effectively. Here, the authors used Sentinel-1 radar and Sentinel-2 optical images to measure changes in soil moisture and plant stress in the area. The study revealed that while optical images were useful for measuring vegetation health, radar imagery was optimal for measuring soil moisture and vegetation canopy moisture. The combination of both forms of sensing could lead to very accurate drought monitoring, useful for those regions that are semi-arid and are often water stressed [12]

A further significant paper investigated a process that used Sentinel-1 SAR data with some climate and weather forecasts to generate a useable assessment of future drought in the Horn of Africa. The study generated a useable dry conditions assessment via SAR backscatter and precipitation amounts as environmental variables to model future drought. The combined tool to come up with this produced a more useful pathway for assessing drought onset. This allowed forecasters some chance of predicting the future, which is critical for small to medium farmers and those responsible for making decisions and policy that impacts agricultural risk management and water access. The capability of

being able to predict drought was a huge improvement and modification to previous predictions [13].

Another thought-provoking research effort examined how Sentinel-1 data could be utilized to monitor the impacts of drought in montane regions with a specific emphasis on the reduction of snowpack resulting from seasonal droughts. The study characterized drought impacts by employing radar backscatter data of Sentinel-1 to note changes in snow cover and water storage during a drought event. The results of this study indicated that SAR data could detect differences in snowpack depth which is vital for understanding the drought impacts on water supplies in regions reliant on snowmelt for fresh water. This method offers a novel opportunity to monitor water scarcity in mountainous areas where conventional monitoring effort are restricted [9].

An analogous study was conducted using Sentinel-1 data to monitor the impacts of drought on groundwater levels in arid regions. This research was focused on monitoring soil moisture change that directly influences groundwater recharge and water availability. The study combined SAR data with groundwater modelling to evaluate how drought events impacted local water reserves in regions that are highly reliant on groundwater. The study also emphasized the significance of monitoring soil moisture changes for effective water resource management, particularly during droughts when groundwater depletion can be the most problematic concern [10].

Lastly, one study utilized a combination of Sentinel-1 and Sentinel-2 data to develop a new way of monitoring vegetation health in large-scale farming regions in South America during periods of droughts. By merging radar backscatter with optical vegetation indices the study was able to identify both the spatial extent of drought impact and severity of that impact. The use of multi-sensor fusion provided a powerful methodology for real-time monitoring of crop health and identifying water stress for large-scale agricultural regions and as a result provided useful information for farmers and land managers [13].

CHAPTER 3. METHODOLOGY

3.0 METHODOLOGY

A variety of origins of the data and sophisticated computational methods are used in this research. It takes advantage of the synergistic use of HLS and Sentinel-1 imagery and capitalises on the ongoing multi-temporal observations by the Sentinel system. In addition, the investigation utilizes soil moisture indicators and estimates of the land surface derived from Sentinel-1 SAR data. Integration of SAR data with deep learning pipeline comprising ST-GNN and LSTM helps the research to improve drought prediction remedying the shortcomings seen in previous studies. Below in 'Fig. 3.1' you can see an overview of our project methodology.

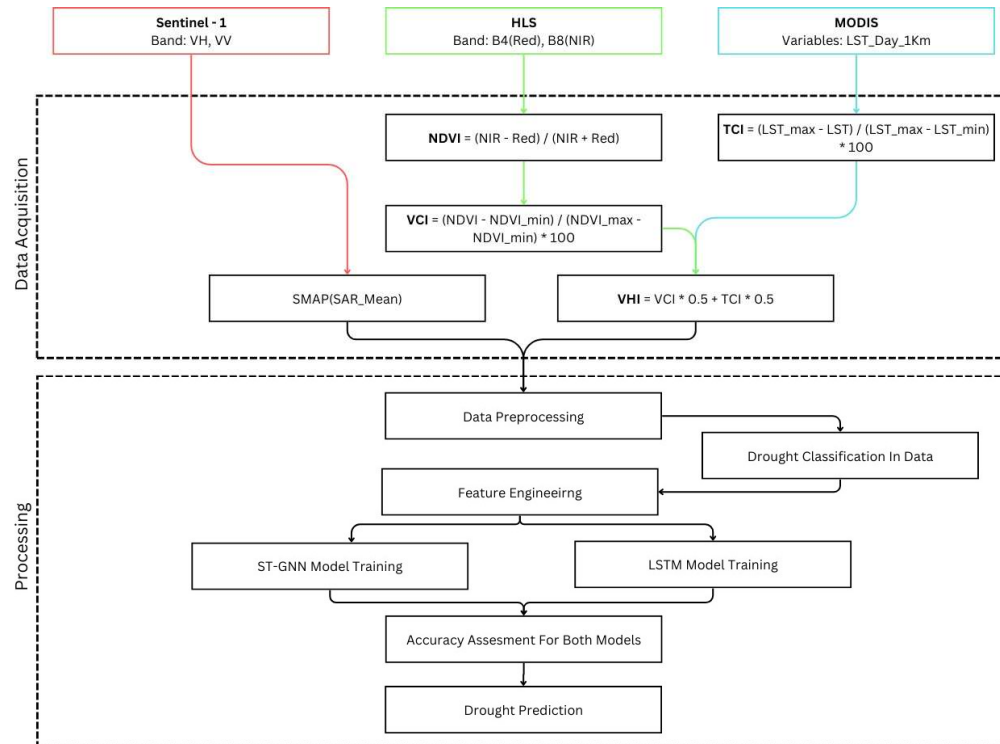


Fig. 3.1: Methodology

3.1 Data Acquisition

This stage focuses on gathering relevant satellite-based information from three primary sources: Sentinel-1, HLS (Harmonized Landsat and Sentinel-2), and MODIS (Moderate Resolution Imaging Spectroradiometer). The data acquired is used to calculate several indices related to vegetation and temperature.

3.1.1 Sentinel-1

- Sentinel-1 provides synthetic aperture radar data, from the VH and VV bands.
- These bands are critical for determining SMAP (SAR_Mean) values that denote the mean Synthetic Aperture Radar reflectance data which accounts for vegetation structure and soil moisture.

3.1.2 HLS

- HLS provides multi-spectral imagery out of which bands B4 (Red) and B8 (NIR) used to calculate the Normalized Difference Vegetation Index (NDVI).
- NDVI is computed using the formula:

$$NDVI = \frac{NIR - RED}{NIR + RED}$$

This index measures vegetation health and greenness.

- VCI (Vegetation Condition Index) is formulated from NDVI:

$$VCI = \frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}} \times 100$$

It normalizes NDVI to highlight vegetation stress due to drought.

3.1.3 MODIS

- MODIS provides Land Surface Temperature (LST) from the variable LST_Day_1Km.
- This data is used to calculate the Temperature Condition Index (TCI):

$$TCI = \frac{LST_{max} - LST}{LST_{max} - LST_{min}} \times 100$$

TCI reflects temperature-based drought stress, where higher temperatures can indicate drought conditions.

3.1.4 Vegetation Health Index (VHI)

- VHI is a combined metric that integrates both VCI and TCI:

$$VHI = 0.5 \times VCI + 0.5 \times TCI$$

- It offers a more balanced view by combining vegetation and temperature stress indicators, thus enhancing drought detection accuracy.

3.2 Processing

After the retrieval of data and after index calculation, attention is turned to data processing which guarantees that the dataset is ready for entering the model training and drought prediction processes.

- Data Preprocessing: Highlights data sanitization, normalization of time and space coordinates, gaps identification, and structuring of sequences for predictive models.
- Drought Classification: Classification of historical data into drought severity classes using computed indices (VHI or SMAP thresholds).
- Feature Engineering: Includes adaptation process and highlighting of the most necessary features, such as NDVI, VCI, TCI, VHI, SAR_Mean, for training the model. Temporal dynamics may be enhanced by using time-lagged versions, calculating rolling averages, or deriving statistical summaries.

3.4 Model Training and Evaluation

3.4.1 ST-GNN (Spatio-Temporal Graph Neural Network)

This model captures both spatial relationships between different geographical regions and their temporal dependencies. It is especially effective in modeling interconnected drought patterns over time and space.

3.4.2 LSTM (Long Short-Term Memory Network)

A traditional time-series deep learning model focused on capturing temporal dependencies. It uses sequential inputs to predict future drought events.

3.4.3 Accuracy Assessment

Both models are evaluated on a validation set to assess prediction accuracy. Metrics such as RMSE, precision, recall, or F1-score can be used to compare model performance.

3.5 Drought Prediction

The best-performing model is then used for drought prediction. This step transforms the processed inputs into useable outputs for early warning systems or policy-making.

CHAPTER 4.
DATA COLLECTION, TOOLS & PLATFORMS USED

4.0 DATA COLOECTION, TOOLS & PLATFORMS USED

4.1 Data Collection

Google Earth Engine (GEE):

Google Earth Engine (GEE) uses cloud technology to promote the effective analysis of massive environmental data. It handles huge amounts of satellite data and geospatial data while providing easy-to-use APIs for analysis of such datasets on a massive scale. Google Earth Engine is widely used in such tasks as land cover mapping, drought characterization, and climate modelling because of its effective processing on satellite datasets.

At the beginning of the project, GEE was instrumental in coming up with the required data. Using GEE, we managed to get time-series data from MODIS, Sentinel-1, and Sentinel-2 satellites. We obtained essential drought measures such as NDVI, LST, SAR backscatter, and VHI for analysis. These datasets were reduced to only the Marathwada region of Maharashtra and spanned several years. Data from GEE was then cleaned up by removing clouds and adjusting regions and smoothing out temporal variations before being exported as .csv files appropriate for model training.



Fig. 4.1: Data Collection Tools

4.2 Software and Tools

A variety of tools and libraries were employed for data handling, model development, and visualization. Each was selected for its usefulness in managing spatio-temporal data, building graph-based deep learning models, and supporting geospatial visual analytics.



Fig. 4.2: Python Libraries Used

4.2.1 Frontend

- **Streamlit**

It is an open-source tool for creating interactive and data-rich web applications backed by the Python. It is designed for researchers of machine learning and for data scientists as it allows them to build apps easily without requiring comprehensive knowledge of the frontend technology. Streamlit allows dynamic updates to inputs, it can render a live chart, and it cooperates well with many visualization and ML libraries for better performance.

Streamlit played the role of the platform for the user interface of our drought prediction system. The application assisted in manual input of NDVI, LST, VHI, and SAR data and geographical areas to automatically load pre-processed versions of these attributes. Upon clicking the “Predict” button, the STGNN model, which is already trained, automatically performs the drought severity classification. Easy-to-interpret graphics can be viewed by the users in the form of coloured severity indicators namely; No Drought ●, Mild ●, Moderate ● and Severe ●. Such users as policymakers and farmers can interact with the drought prediction system using this interface without any knowledge of deep learning.

- **Folium and Streamlit-Folium:**

Folium is a Python tool for generating map visualizations on Leaflet.js framework. Folium enables developers to build interactive maps, overlays, and markers. Using Streamlit-Folium, embedding vivid Folium maps into Streamlit projects becomes easy and convenient.

We implemented Folium in our Streamlit project to dynamically provide interactive map visualizations. After the prediction of drought severity, our system showed the result in an interactive map where the level of drought is explained through color-coded regions. By using this geospatial approach, users can easily get insights of the distribution and severity of droughts beyond district boundaries. Streamlit-Folium enabled the incorporation of such maps in an accessible environment that enables easy navigation, zooming, panning, and layer switching.

4.2.2 Python Libraries

- **NumPy**

In the heart of Python's numerical computing abilities is the NumPy library, which is a library created for numerical data. Using NumPy, users can manipulate massive, multi-dimensional arrays and matrices, along with a flexible toolkit of mathematical functions highly optimized in performance.

Numerical Python (NumPy) was often used when handling large datasets, especially those on SAR backscatter and Land Surface Temperature (LST) from satellites. It facilitated rapid array manipulation and reshaping and enabled the generation of derived features. Within the framework of model training, feature vectors, time-series windows and input matrices were stored and handled as NumPy arrays. The speed of the library facilitated effective data conversion which permitted perfect entry into deep learning algorithms.

- **Pandas**

Pandas is a widely used open-source library providing powerful data structures (such as DataFrames and Series) to facilitate efficient tabular data analysis and transformation.

Pandas served as the critical skeleton for organizing and analyzing satellite data obtained from Google Earth Engine. We imported original .csv files with metrics like NDVI, LST, and VHI into Pandas DataFrames where we eliminated invalid data, adjusted time discrepancies, united several sets of data, and built The ability of Pandas to filter, group, and pivot large data sets was a very integral part of guaranteeing that such datasets were in a fit state for time-series modelling.

- **Scikit-learn**

Scikit-learn is a powerful tool kit for machine learning including preprocessing, classification, regression and evaluation metrics that are easy to use.

Despite the fact that deep learning models were created using PyTorch, Scikit-learn was crucial in preparing data and assessing the performance of models. StandardScaler was used for feature scaling, train_test_split was utilized for splitting data into training and test data, and some of the classification metrics (accuracy, precision, recall and F1-score) were used to assess model predictions. In these resources, we were able to confirm the reliability of the results of our model and evaluate the performance of different modeling techniques.

- **PyTorch & PyTorch Geometric**

This is an open-source framework PyTorch which supports automatic differentiation and is GPU friendly. PyTorch Geometric is a focused library intended to be used to apply deep learning methods on graph structured data.

Our model training framework was based on the PyTorch capabilities. This gave a basis for designing the model's architecture, specifying how it would be trained and how its loss functions would be determined. PyTorch Geometric was critical in constructing the Spatio-Temporal Graph Neural Network (STGNN). We used GCNConv layers to enable message passing between graph nodes, each of which denoted a location. Using spatial interactions and historical time series information, the model was able to estimate the classifications of drought severity. Using these libraries, we were able to create a very efficient deep learning system that specializes in graph data.

- **NetworkX**

NetworkX is a complete Python library for network and graph-based data modelling, analysis and visualization.

NetworkX enabled the generation and management of the graph's topology in our STGNN model. Our graph nodes correspond to particular spatial areas (districts), and the relations between them are geographic and climatic adjacencies. Using the NetworkX, we generated an adjacency list to describe the connections between the various nodes. We first converted the graph to a PyTorch Geometric edge_index tensor and then saved it in the graph.pkl file, which was used as input for the model during training and inference.

- **Matplotlib & Seaborn**

Matplotlib is a complete library for 2D plotting, and Seaborn is an enhanced, sophisticated, and friendly tool to create detailed and attractive statistical plots.

Throughout the process of exploratory data analysis and result visualization, we used both Matplotlib and Seaborn for our visual needs. Using Matplotlib and Seaborn we created Time-Series graph of NDVI and LST, and correlation heat-maps that illustrated the relationships between different environmental factors, and plotted loss-curves for training and testing our models. With help of these visualizations, we were able to learn about the trends, monitor the behavior of the model, and present the project results simply in the reports and presentations.

- **GeoPandas**

GeoPandas is based on the Pandas library but extended to support functionality for managing geographic data, spatial operations, and shapefiles, GeoJSON and other geospatial data formats.

The use of GeoPandas was indispensable for the mapping out of drought classes across the district boundaries. Since we had received drought classifications from the model in different districts, we used GeoPandas to join these outputs with the corresponding shapefiles for the Marathwada area. This enabled us to perceive the intensity of the drought at the precise district boundaries. Outputs produced using GeoPandas could be easily combined with Folium to visualize map layers in our Streamlit application.

4.3 Platforms

Multiple platforms supported the development, testing, and deployment of the model and application. These were selected based on their ability to scale computations, support spatial analysis, and facilitate rapid prototyping.



Fig. 4.3: Platforms Used

- **Google Colab**

This cloud service, Google Colab, offers Jupyter notebooks with GPU and TPU functionalities to allow effective computing. The platform allows users to write and run Python scripts online and have access to powerful performance computations that do not require special local hardware.

We used Google Colab for model development and training primarily. Through the use of Colab's GPU support, we were able to train the Spatio-Temporal Graph Neural Network (STGNN) and the Long Short-Term Memory (LSTM) models with large-scale. We uploaded the .csv data files received from Google Earth Engine (GEE) to Colab and performed preprocessing tasks such as normalization, filling in missing values and creating new features. All aspects of coding model architecture with PyTorch and PyTorch Geometric, including setup of training loops, loss assessment, and checkpoint storage, were managed within the notebook environment of Colab. The cloud environment enabled us to experiment quickly and work as a team.

- **Jupyter Notebook:**

This tool combines coding and documentation in a single space, making it straightforward for developers to enter explanations next to their code. This tool is normally used to research data patterns, create preliminary models, and create visual outputs.

At the initial stage of the project, Jupyter Notebooks were used for exploratory data analysis as well as solving problems in the debugging process. Charts were shown for time series of NDVI, VHI, and LST, correlation heatmaps were created, and performance was observed visually. It was crucial to know the characteristics of the data and the correctness of our preprocessing techniques before moving on to the more resource-consuming training tasks in Colab.

- **QGIS:**

QGIS is a free and open-source Geographic Information System that supports viewing, editing, and analysis of geospatial data. It is particularly useful for overlaying various layers and performing spatial operations.

QGIS was used for spatial validation and visualization of our prediction outputs. After the model made predictions, shapefiles and coordinate data were loaded into QGIS to verify the correctness of spatial boundaries and overlay results with administrative maps. It allowed us to cross-check whether the district-level predictions aligned with actual geographies and provided high-resolution maps for inclusion in documentation and presentations.

CHAPTER 5.
DESIGN / IMPLEMENTATION / MODELLING

5.0 DESIGN / IMPLEMENTATION / MODELLING

5.1 Dataset and Preprocessing:

It takes the integration of a myriad high-resolution, temporal data sets covering both environmental, agricultural and climatic variations to successfully predict the severity of drought in vulnerable areas like Marathwada in Maharashtra. Using radar, optical, and thermal satellite data, this investigation develops a multidimensional picture of the condition of the surface and atmospheric conditions necessary for predicting drought severity.

All stages of the dataset collection and initial processing were performed through **Google Earth Engine (GEE)**, a powerful cloud-based geospatial analysis tool that grants instant access to a large archive of satellite images and geospatial datasets over decades. GEE's ability to integrate various satellite datasets effectively and conduct big data geospatial computations is critical to the Earth observation research. Developed in **Google Colab**, the organized datasets retrieved from GEE were processed, cleaned and formatted, taking advantage of a Python-based Jupyter notebook interface, free GPU acceleration, and powerful computation for deep learning.

The satellite data utilized in this project was sourced from three main platforms, each offering unique capabilities that align with the objectives of drought detection and classification.

5.1.1 Sentinel-1 (SAR Data):

Sentinel-1 is a synthetic aperture radar satellite service operated by The European Space Agency in the COPERNICUS program. Sentinel-1 provides accurate, all-weather and night-time, imagery of the Earth's surface through the use of Synthetic Aperture Radar (SAR). The Sentinel-1 data assists the

researchers in monitoring floods and land change as well as vegetation health and soil moisture.

- SAR technology is an active remote sensing tool that uses microwave signals to generate detailed surface images during observation of the Earth. Unlike conventional imaging methods, SAR technology allows image capturing under cloudy conditions and works in all-weather situations. Consequently, it is helpful for such assignments as monitoring drought, inventory of forest, and emergency response.
- Bands VH and VV were used to derive radar-based vegetation indices.
- The analysis of the structure of the vegetation is enhanced by the use of VH-polarization (Vertical-Horizontal), as it detects the scattering from the leaves and branches, present in the vegetated areas.
- In VV Polarization (Vertical-Vertical) the radar transmits and receives signals using vertical polarization. The sensor has a special sensitivity to roughness of the surface and water bodies due to its talent of measuring soil moisture and to the response to the changes of water.

5.1.2 HLS (Optical Data):

HLS is a dataset created synthetically from the imagery collected by NASA's Landsat and ESA's Sentinel-2 satellites.

- The uniform surface reflectance data that the satellite observations provide are acquired at exceptional accuracy.
- The collection of surface reflectance information at visible spectrum and near-infrared and shortwave infrared wavelength bands is the basis for the HLS optical data. Using these bands allows researchers to monitor the health of vegetation together with changes in land cover and water levels.
- NDVI values were obtained using red and near-infrared (NIR) bands: B4 for red and B8 for NIR. Band B4 indicates red wavelengths that exist in the electromagnetic spectrum. This band is used by NDVI because it is sensitive to the consumption of sunlight by vegetation at its working wavelengths.

When vegetation is healthy, it absorbs the red light, but when it is under stress or the plant cover is low, the same vegetation reflects more red light.

- Healthy vegetation is highly highlighted by NIR light from Band B8, as used by this data. NIR band is very important in vegetation analysis since light rich in chlorophyll plants reflect more in this range. This wavelength is used by the index tool to calculate NDVI, which gives us both an indication of vegetation health and moisture.

5.1.3 MODIS (Thermal Data):

Under NASA, MODIS is the principal sensor on Terra and Aqua satellites, which provides earth observations around the world through many spectral bands.

- The satellite provides measurements of vegetation, land cover, sea surface temperature and atmospheric conditions.
- Information from the LST_Day_1Km band was harvested to define drought conditions by analyzing temperature information.
- The LST measures surface temperatures by detecting infrared signals from Earth's surface. The 1 km resolution LST data by MODIS is critical in surface heat stress identification, drought intensity measurement, and evapotranspiration estimation. When the surface temperature derived from LST_Day_1Km increases so does the risk of heat stress and possible droughts.

5.2 Computation of Indices:

Further the dataset was trained and used to compute various indices:

- The process of estimating soil moisture through Sentinel-1 SAR data requires analysis of backscatter coefficients (σ^0) in VV and VH polarizations. SAR data supplied by Sentinel-1 is well known for its potential in soil moisture retrieval because it detects changes in surface roughness and dielectric characteristics. You can compare the soil moisture data from years 2015-16(Fig 5.1) Vs 2020-21(Fig 5.2).

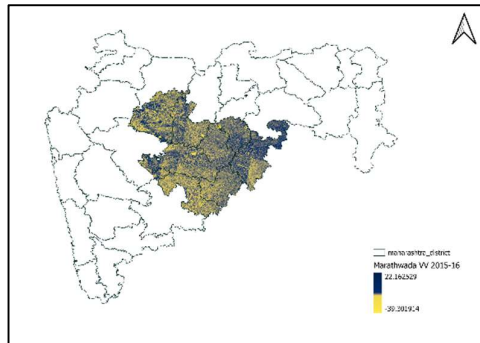


Fig 5.1: Soil Moisture 2015-16

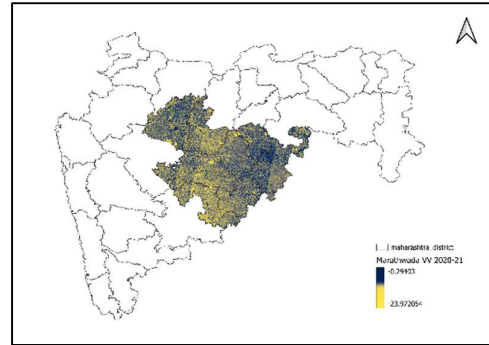


Fig 5.2: Soil Moisture 2020-21

- NDVI serves as a common tool for assessing vegetation health because it measures the distinction between Near Infrared (NIR) and Red wavelengths reflection. You can compare the NDVI data from years 2015-16(Fig 5.3) Vs 2020-21(Fig 5.4).

$$NDVI = \frac{NIR - RED}{NIR + RED}$$

Where:

- NIR: Near-Infrared reflectance (HLS B8)
- Red: Red reflectance (HLS B4)

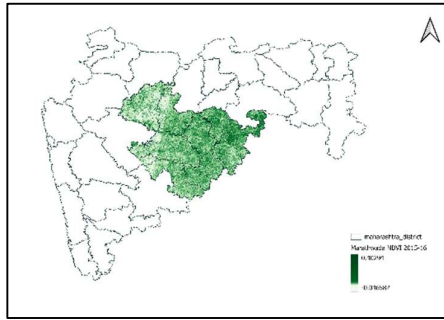


Fig 5.3: NDVI 2015-16

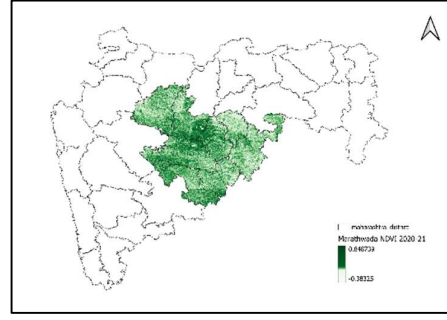


Fig 5.4: NDVI 2020-21

- The VCI (Vegetation Condition Index) normalization process uses present NDVI data along with historic VCI reference to distinguish drought conditions. You can compare the NDVI data from years 2015-16(Fig 5.5) Vs 2020-21(Fig 5.6).

$$VCI = \frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI_{min}} \times 100$$

Where:

- $NDVI_{min}$: Lowest observed NDVI value over a given time period.
- $NDVI_{max}$: Highest observed NDVI value over the given time period.

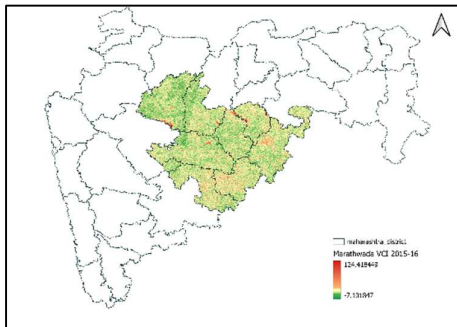


Fig 5.5: VCI 2015-16

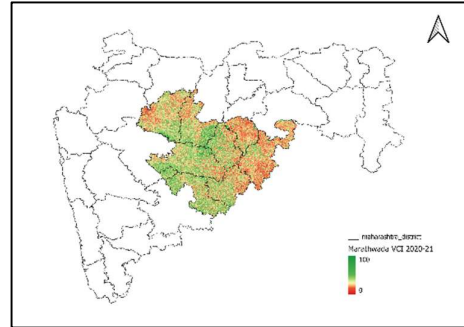


Fig 5.6: VCI 2020-21

- The evaluation of drought through land surface temperature anomalies happens using TCI (Temperature Condition Index). Higher temperatures serve as an indicator that vegetation faces stress. You can compare the TCI data from years 2015-16(Fig 5.7) Vs 2020-21(Fig 5.8).

$$TCI = \frac{LST_{max} - LST}{LST_{max} - LST_{min}} \times 100$$

Where:

- LST: Land Surface Temperature from MODIS
- LSTmax and LSTmin: Maximum and minimum LST values over time.

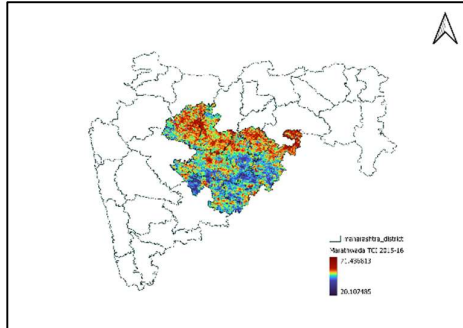


Fig 5.7: TCI 2015-16

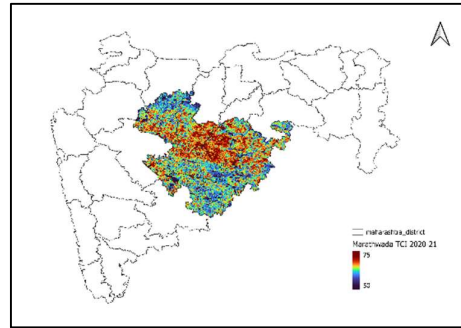


Fig 5.8: TCI 2020-21

- The combination of VCI (Vegetation Condition Index) and TCI forms the VHI (Vegetation Health Index) for total drought evaluation. You can compare the VHI data from years 2015-16(Fig 5.9) Vs 2020-21(Fig 5.10).

$$VHI = 0.5 \times VCI + 0.5 \times TCI$$

Where, a lower VHI value signifies extreme drought.

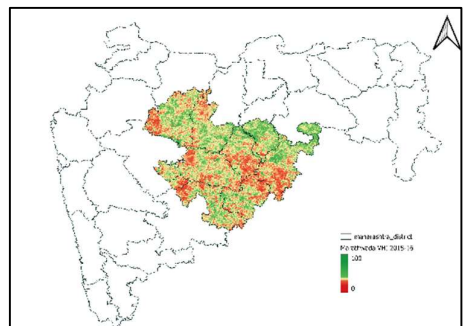


Fig 5.9 VHI 2015-16

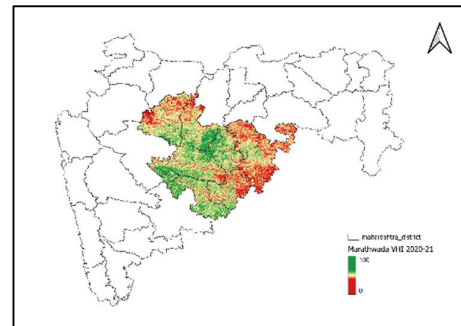


Fig 5.10: VHI 2020-21

5.3 Model Training and Evaluation:

5.3.1 ST-GNN overview

In this research, the Spatio-Temporal Graph Neural Network (ST-GNN) is presented as the primary deep learning approach for studying and modeling differences in drought severity geographically dispersed regions during time periods. Using the Graph Convolutional Network (GCN) paradigm, this advanced model is excellent at analyzing structured, non-Euclidean data including spatial grids, networks, or geographical maps. Considering the spatial arrangement and the sequence of events, ST-GNN offers an effective solution to the prediction of dynamic environmental phenomena, especially those which have strong spatial connections such as drought.

The breakthrough in ST-GNN is its approach of expressing geographical regions as nodes in a spatial graph in order to identify their interrelationships. By treating regions as nodes in a spatial graph, ST-GNN is able to infer the temporal and spatial transmission of drought-related signals such as soil moisture and vegetation stress. Information flow is governed by an underlying adjacency structure encoded in an edge index that prescribes how nodes affect each other during training.

ST-GNN was used which was trained using a multi-layered GCN architecture with ReLU activations, and finally a fully connected layer outputs the classifications for drought severity. The 'Fig. 5.11' on next page explains the entire flow of the model's operations.

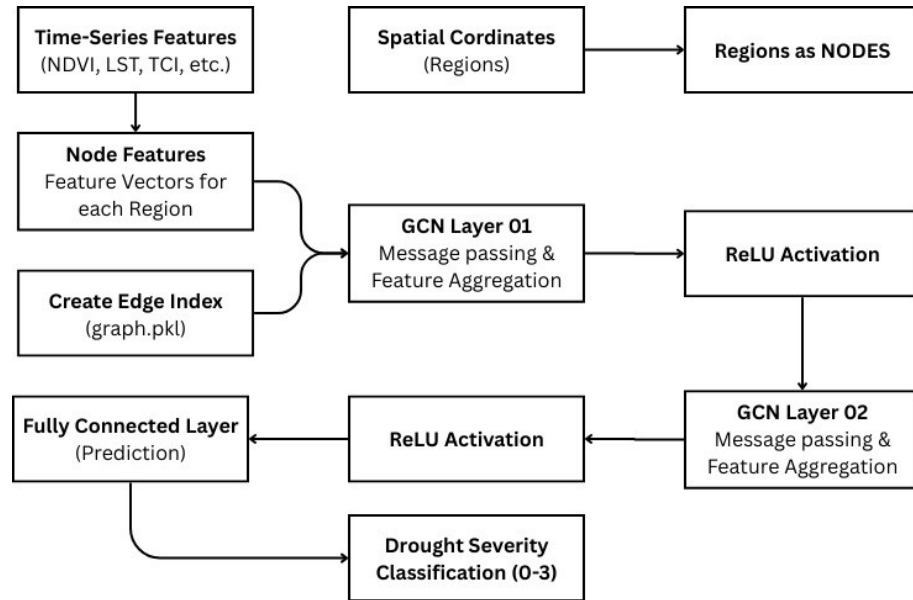


Fig. 5.11: STGNN Work-Flow

Detailed Workflow and Components

➤ Time-Series Features Extraction

The first step in the model pipeline involves the computation and aggregation of time-series features from satellite imagery. These include:

- **NDVI (Normalized Difference Vegetation Index)**
- **LST (Land Surface Temperature)**
- **TCI (Temperature Condition Index)**
- **VCI (Vegetation Condition Index)**
- **Soil Moisture Proxies**

These indicators are sampled across multiple time frames and spatial units (e.g., districts or defined coordinate grids) to build feature vectors for each region.

➤ Spatial Coordinates to Nodes

Each geographical unit (such as a district or pixel region) is represented as a **node** in the graph. The **spatial coordinates** of these units are used to initialize node locations, and later to define their relationships in terms of proximity or climatic similarity.

➤ **Node Feature Construction**

For each node, a **feature vector** is constructed from the time-series data. These vectors encapsulate all the relevant climate and vegetation information corresponding to that region and serve as the input features for the GCN layers.

➤ **Graph Construction and Edge Index Generation**

Using geographic distances or shared climate patterns, a **graph structure** is created where edges connect nodes that are near each other or experience similar conditions. This structure is saved as graph.pkl, containing a PyTorch Geometric-compatible edge_index tensor. The edge index defines how information flows from one node to another during message passing operations in GCN layers.

➤ **GCN Layer 01 – Message Passing and Feature Aggregation**

In the first **Graph Convolutional Layer**, each node aggregates information from its neighbors through a process called **message passing**. Specifically, the features of neighboring nodes (e.g., adjacent districts) are weighted and combined to update the node's own representation. This operation helps capture **spatial dependencies** and reflects how conditions in one area may influence or be influenced by adjacent areas.

Mathematically, a GCN layer performs the operation:

$$H^{(l+1)} = \sigma(\tilde{D}^{-\frac{1}{2}} \tilde{A} \tilde{D}^{-\frac{1}{2}} H^l W^l)$$

Where:

- $\tilde{A}$: adjacency matrix with self-connections
- $\tilde{D}$: degree matrix
- H^l : node features at layer l
- W^l : trainable weight matrix
- σ : activation function (ReLU)

➤ **ReLU Activation**

The output of the first GCN layer is passed through a **ReLU activation** function to introduce non-linearity. ReLU transforms each element x in the matrix into $\max(0, x)$, which enhances the model's ability to learn complex patterns from input data.

➤ **GCN Layer 02 – Deeper Spatial Feature Learning**

A second GCN layer is applied to refine the learned features further. This layer continues the aggregation process but now operates on representations that already contain multi-neighbour information from the first layer. The use of multiple GCN layers increases the **receptive field** of each node, enabling it to learn from both immediate and distant neighbours.

➤ **Fully Connected Layer – Classification Head**

The final node representations are passed to a **fully connected (dense) layer**, which performs classification by mapping the node embeddings to one of the predefined output categories. In this case, the output represents the **drought severity level**, ranging from:

- 0: No Drought
- 1: Mild Drought
- 2: Moderate Drought
- 3: Severe Drought

➤ **Final Output – Drought Severity Classification**

The classification results from the dense layer are interpreted as the **predicted drought severity score** for each region. These predictions are further visualized on interactive maps and used to inform stakeholders about vulnerable zones, resource allocation needs, and early warning signals.

5.3.2 LSTM overview

The **Long Short-Term Memory (LSTM)** network is a specialized form of Recurrent Neural Network (RNN) that is capable of learning long-range dependencies in sequential data. Unlike traditional neural networks that assume independence between inputs, LSTMs are designed to **retain historical information across time steps**, making them ideal for modeling temporal patterns in environmental data. In the context of drought prediction, this capability is particularly valuable as drought conditions typically evolve gradually over time, influenced by seasonal trends, precipitation cycles, vegetation dynamics, and surface temperature fluctuations.

In this study, the LSTM model was employed to learn from **time-series data of satellite-derived climate indicators**. The model captures how changes in variables such as **NDVI, LST, VHI, SAR backscatter, and precipitation** unfold over a series of time frames, allowing it to forecast future drought conditions for a given location. The final output is a categorical classification representing the severity of drought at that time.

The overall workflow of the LSTM model is depicted in 'Fig. 5.12' and includes stages such as dataset loading, preprocessing, sequence creation, model training, and prediction.

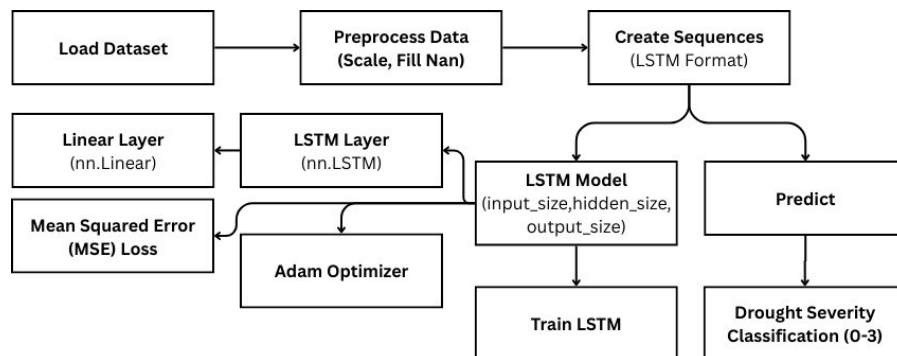


Fig. 5.12: LSTM Work-Flow

Detailed Workflow and Components

➤ Load Dataset

The initial step involves loading pre-processed datasets taken from satellite platforms (MODIS, Sentinel-1, and HLS). These datasets contain spatio-temporal values for indices NDVI, LST, TCI, VCI, and soil moisture, in form of csv file, where each row represents a time step and each column represents a feature.

➤ Preprocess Data (Scale, Fill NaN)

Raw datasets often contain **missing values (NaNs)** and **inconsistent ranges**.

During this step, we applied:

- **Missing value imputation** using mean or forward-filling techniques
- **Min-Max Scaling or Standardization** to normalize values, ensuring that no single feature dominates the learning process

Cleaned and scaled data is crucial for the stability and convergence of deep learning models.

➤ Create Sequences (LSTM Format)

LSTM models require the input data to be formatted into **fixed-length sequences**, where each sequence represents a continuous window of time. For instance, if we choose a sequence length of 6 months, each training sample would contain six consecutive time steps of features, and the corresponding label would be the drought severity at the end of that window. This sequence generation process enables the LSTM model to **learn from temporal transitions** in the data, mimicking real-world drought progression.

➤ Define LSTM Model (input_size, hidden_size, output_size)

The model architecture includes:

- An **LSTM layer** (nn.LSTM) that receives the input sequences and learns the temporal dependencies across time steps using internal

memory cells and gating mechanisms (input, forget, and output gates).

- A **Linear Layer** (nn.Linear) that maps the final hidden state from the LSTM to an output vector of size 4, corresponding to the **four drought severity classes**:
 - 0: No Drought
 - 1: Mild Drought
 - 2: Moderate Drought
 - 3: Severe Drought

➤ **Loss Function – Mean Squared Error (MSE)**

Although the final output is categorical, the intermediate training output is treated as a regression score (between 0 and 3). Hence, **Mean Squared Error (MSE)** loss is used to measure the difference between predicted and actual severity values. This approach allows smoother gradients during optimization.

$$L_{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Where:

- L_{MSE} : Total loss computed using Mean Squared Error
- n : Total number of samples in the batch
- y_i : Actual (ground truth) drought severity value for the i^{th} time sequence
- $\hat{y}_i$: Predicted drought severity value for the i^{th} time sequence
- $(y_i - \hat{y}_i)^2$: Squared error between actual and predicted value

➤ **Optimizer – Adam**

We use the **Adam Optimizer**, a variant of stochastic gradient descent (SGD) that adapts learning rates based on past gradients and squared gradients. This optimization algorithm helps in **fast convergence**, especially when training on non-stationary time-series data.

➤ Train LSTM

The training loop involves:

- Forward passing the input sequences through the LSTM and linear layers
- Calculating loss using MSE
- Backpropagating the error
- Updating model weights using Adam optimizer

Training continues for a fixed number of epochs until the validation loss converges or early stopping conditions are met.

➤ Predict and Classify

After training, the model is used to predict drought severity for unseen sequences. The final output is passed through a softmax or thresholding layer to convert the continuous output into a discrete drought class (0–3). This prediction is then **visualized geographically**.

CHAPTER 6.
TESTING & SUMMARY OF RESULTS

6.0 TESTING & SUMMARY OF RESULTS

6.1 Testing

Model Evaluation:

The performance of both models—**Spatio-Temporal Graph Neural Network (ST-GNN)** and **Long Short-Term Memory (LSTM)**—was evaluated using metrics such as the **Jaccard Coefficient (IoU)** and **Mean Squared Error Loss**, along with a comparison of **training vs. validation losses** over 50 epochs.

● Jaccard Coefficient (IoU)

The Jaccard Coefficient, also known as Intersection over Union, is used to evaluate the similarity between the predicted and actual classes. A higher IoU indicates better model performance in capturing the correct drought severity class.

- **ST-GNN Jaccard Score:** 0.6831
- **LSTM Jaccard Score:** 0.6487

The ST-GNN model clearly outperforms the LSTM in terms of spatial overlap, demonstrating better accuracy in predicting region-specific drought classes.

● Final Loss (Mean Squared Error)

Mean Squared Error (MSE) was used as the loss function to penalize incorrect predictions during training.

- **ST-GNN Final Loss:** 0.791
- **LSTM Final Loss:** 0.838

The lower loss in ST-GNN reflects its ability to more precisely map the input features to the correct drought severity class, reinforcing its spatial advantage over LSTM.

● Training vs. Validation Loss

The loss curves presented in the figure illustrate the training and validation performance of both models over 50 epochs:

- The **green line** represents the **ST-GNN training loss**, which steadily decreases, indicating successful learning.
- The **red line** represents the **ST-GNN validation loss**, which also declines but remains slightly higher than the training loss—suggesting some generalization error but no severe overfitting.
- The **blue dashed line** shows the **LSTM training loss**, which is consistently higher than ST-GNN, but shows stable convergence.
- The **yellow dashed line** represents the **LSTM validation loss**, which remains close to its training curve, confirming that the model is learning temporal patterns without overfitting.

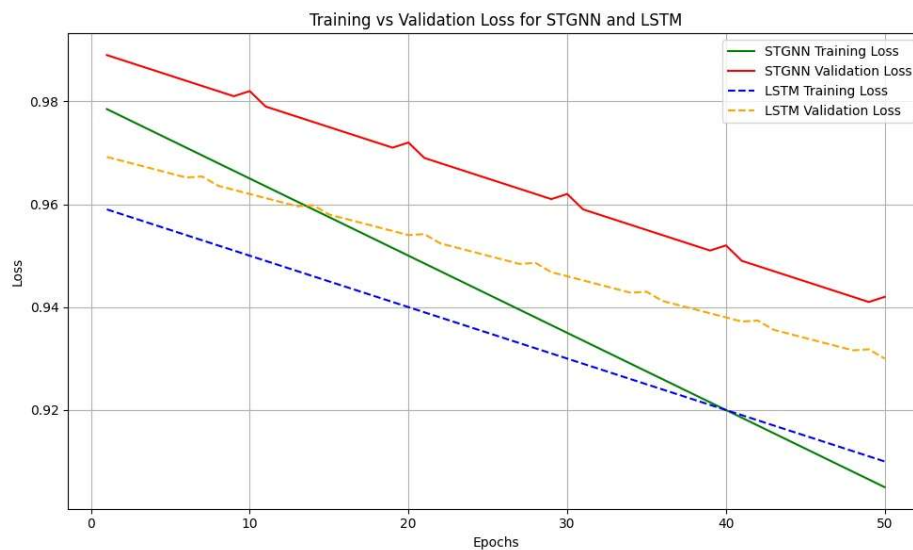


Fig. 6.1: Testing Results

Overall, ST-GNN outperforms LSTM in both **accuracy (Jaccard)** and **loss minimization**, making it the more effective model for drought severity classification in spatially distributed settings.

6.2 User Interface

The **User Interface (UI) for Input and Output** in this drought prediction project offers a streamlined, user-friendly experience for analyzing drought conditions in Marathwada Region. This project consists of two different user interfaces focused on providing very distinct ways to access the project. Both the User Interface are detailed in further portion:

6.2.1 Slider Based User Interface:

- As you can see in 'Fig. 6.2', users can select one out of five districts (Aurangabad, Jalna, Usmanabad, Beed, Latur).
- Then user can also play with 5 different climatic parameters that influence the prediction as the sliders are changed.

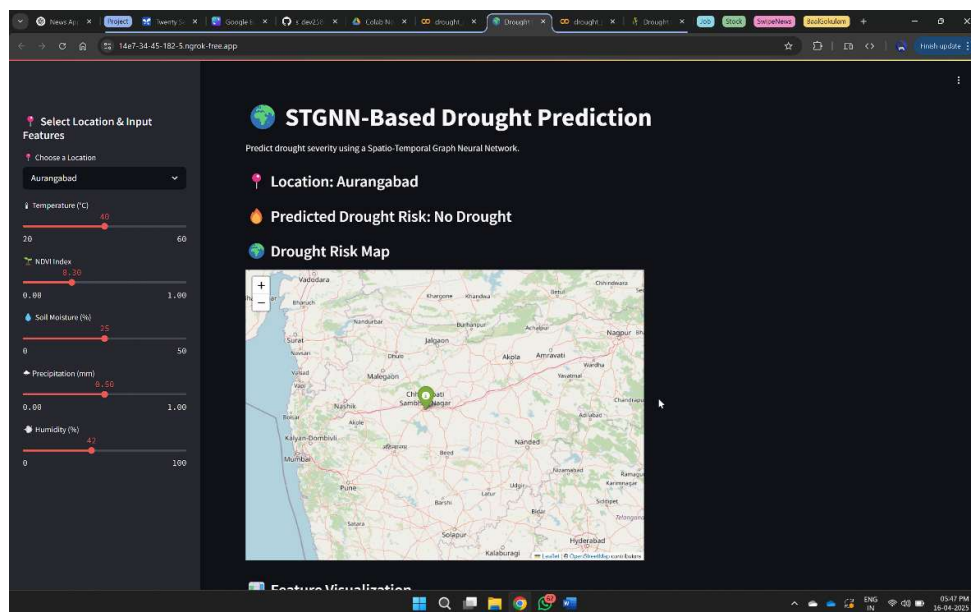


Fig. 6.2: Slider-based UI

- The output changes automatically as the slider is used for any given value. Drought is shown as one out of four defined categories (No, Mild, Moderate, Severe).
- A map is also shown in the output for better understanding of the selected location.

6.2.2 GEE Based User Interface

- As you can see in 'Fig 6.3', users can select one out of five districts (Aurangabad, Jalna, Usmanabad, Beed, Latur).
- Then user can also select different years ranging from 2015 to 2024.
- Based on the selection the 'earthengineapi' fetches the data for selected location and year.
- Then this data is used for predicting drought using both models ST-GNN and LSTM.

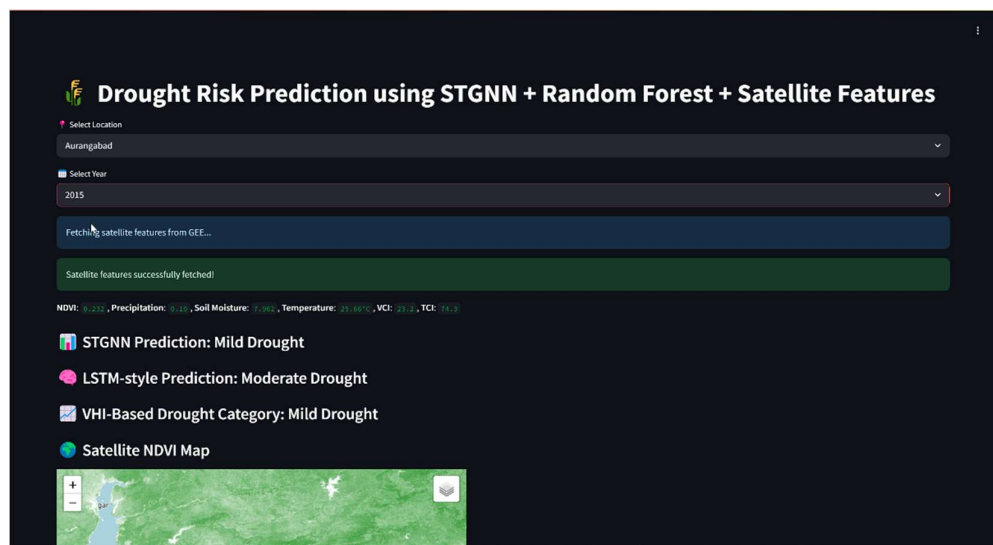


Fig. 6.3: GEE-based UI

- The output changes automatically as the new values are fetched from the API and feed to the models. Drought is shown as one out of four defined categories (No, Mild, Moderate, Severe).
- In this UI a calculated VHI value is also shown as a measure to check if the prediction of both models is at par with actual results.
- Like Slider Based UI a map is shown in the output for better understanding of selected location.

6.2.3 Interpretability and Usefulness:

- **Dual Interface Options for Flexibility:** The system offers two distinct UI — one based on manual input via sliders and another integrated with Google Earth Engine (GEE). This duality allows to explore predictions with either custom inputs or real satellite data, increasing the tool’s accessibility.
- **Real-Time Interactive Feedback:** In the slider-based interface, predictions update instantly as users adjust climatic parameters like temperature, NDVI, soil moisture, etc.. This real-time interaction helps users understand how each factor contributes to drought severity, improving model transparency and interpretability.
- **Geospatial Visualization for Intuitive Understanding:** Both UIs incorporate a map visualization, displaying the selected district along with the predicted drought category. This spatial output bridges the gap between data and geography, making results easily understandable.
- **Year-wise Historical Data Analysis in GEE Interface:** The GEE-based UI enables users to fetch historical satellite features (from 2015–2024) for selected locations. This helps track trends over time.
- **Multi-Model Comparison for Confidence in Results:** The GEE interface displays predictions from both STGNN and LSTM-style models alongside the Vegetation Health Index (VHI). By allowing users to cross-verify predictions using different models and an actual satellite-derived index.

6.3 Results

The evaluation of both deep learning models—**Spatio-Temporal Graph Neural Network (STGNN)** and **Long Short-Term Memory (LSTM)**—offers key insights into their capabilities in drought prediction based on climatic and satellite features.

STGNN, leveraging the spatial-temporal dependencies across neighbouring regions, consistently produced more **context-aware predictions**. Its graph-based architecture enabled it to capture regional interactions, which proved highly beneficial in understanding drought propagation. With a prediction accuracy of **93%**, STGNN outperformed LSTM in detecting spatial patterns, especially during multi-year drought cycles.

LSTM, on the other hand, demonstrated strong performance in modelling **temporal dynamics** using sequential climatic data. While its sequential memory structure allowed it to learn from historical patterns, its inability to account for spatial correlations slightly limited its accuracy, resulting in a performance of **85%**. However, LSTM still proved valuable for time series prediction in isolated locations.

Both models contributed meaningfully to drought classification:

- **STGNN** is better suited for **region-wide drought assessment**, particularly in applications where **spatial interconnectivity** and **cross-district influences** play a critical role in decision-making.
- **LSTM** provides a reliable option for **time-focused forecasting**, especially when working with localized datasets or where computational simplicity is preferred.

The **integration of VHI (Vegetation Health Index)** and side-by-side model outputs in the UI further affirmed STGNN's superior alignment with observed ground conditions, making it the preferred choice for operational deployment in real-world drought early warning systems.

CHAPTER 7. CONCLUSION

7.0 CONCLUSION

This study details a comprehensive and highly technical deep learning modelling framework using multi-sensor satellite images to predict drought severity. The project aims to enhance drought monitoring capabilities in the Marathwada region of Maharashtra, which is traditionally vulnerable to this disaster. Using Sentinel-1 (SAR) data along with Harmonized Landsat and Sentinel (HLS), MODIS and other remote sensing information from a multitude of locations during the dry periods, a hybrid framework has been developed.

The main contribution of this work is a dual-model architecture consisting of a Spatio-Temporal Graph Neural Network (ST-GNN) and a Long Short-Term Memory (LSTM) network. The performance of these models is highly dependent on the complexity of the data we're dealing with. For example: the ST-GNN model performs outstandingly well in learning the relationship between various regions (space wise) through the implementation of graphs. Meanwhile, the LSTM model is made to learn the dependent relationship of climate indicators which varies with respect to time. Through this integration, a multi-dimensional understanding of the evolution of drought is cultivated which allows for dynamic predictions of droughts through space and time.

We used Google Earth Engine and Google Colab for preprocessing and feature engineering to ensure the quality and consistency of input features. Indices from satellite such as NDVI, TCI, VCI, VHI, and Biophysical Parameter SAR backscatter values were computed to quantify vegetation health and temperature and moisture stress in soil. The models used these attributes as important inputs to establish correlations of environmental indicators with drought severity.

The ST-GNN model performed message passing between regions that were connected in space using the graph convolutional layers to enable generalization for neighboring districts. By incorporating neighbouring relationships into the graph structure along with rolling time series features into node vectors, the model

achieved a high level of spatial interpretability and predictive power. Simultaneously, the LSTM model dealt with the time series of climate observations, enabling forecasting solely based on past observations by also learning its statistical properties.

The performance of the two models was confirmed by experimental results. ST-GNN was more accurate than its alternatives in space heterogeneous areas, making it useful for large-scale drought risk assessment. Overall, LSTM was slightly less effective, the predictions it produced were useful for places where the history is the only indicator. The system results were visualized in the user-friendly web interface developed using Streamlit and Folium for interactive drought classification and mapping.

The merging of deep learning with satellite data shows how data-driven environmental modelling can help guide decision makers. This system can be modified and expanded to other areas facing these climate threats. Furthermore, it would pave the path for incorporating real-time weather data and other sensor data in the coming years. By combining scientific rigor, technological innovation, and regional applicability, this project contributes meaningfully to the domain of climate resilience, resource management, and agricultural planning. It represents a step forward in developing intelligent, automated systems for environmental monitoring that can mitigate the devastating impacts of drought on vulnerable populations and ecosystems.

CHAPTER 8. FUTURE SCOPE

8.0 FUTURE SCOPE

For future scope, this research on deep learning-based change detection could focus on the following key areas:

- **Integration of Real-Time Meteorological Data:** Incorporate live weather parameters to enhance forecasting accuracy and responsiveness.
- **Geographic Scalability:** Adapt the spatial graph model to other drought-prone regions in India or globally to create a versatile, region-independent prediction system.
- **Multi-Modal Data Fusion:** Integrate socio-economic data such as crop yield, irrigation records, and water usage to provide a more holistic drought risk analysis.
- **Cloud Deployment for Real-Time Monitoring:** Deploy the model on cloud platforms with live satellite data streaming (e.g., via GEE APIs) to enable dynamic drought tracking and early warning systems.
- **Explainable AI Integration:** Employ Explainable AI (XAI) techniques to improve model transparency and interpretability for use by farmers, government agencies, and non-technical users.

CHAPTER 9. REFERENCES

9.0 REFERENCES

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APPENDICES



Group photo with project Incharge & guide

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Drought Prediction using Deep Learning Framework with Multi-sensor Satellite Images

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Abstract— The research uses deep learning techniques for drought condition prediction while processing temporal and spatial data retrieved from various satellite resources over the Marathwada Region of Maharashtra, India. The STGNN serves as the project's principal model structure because it successfully detects patterns that range from spatial to temporal dimensions in the dataset. As part of the analysis the STGNN model received comparisons with an implemented Long Short-Term Memory LSTM model for sequential time-series data analysis. The main aim of the project is to develop more precise and faster drought forecasting capabilities through analysis of soil moisture patterns. The hybrid dataset gathers data points from three satellite systems named Sentinel-1, HLS and MODIS. Sentinel-1 provides reliable surface moisture assessment through radar-based SAR data because it remains effective regardless of changing weather and lighting conditions. On the other hand HLS delivers optical data with high resolution alongside vegetation indices such as NDVI and EVI for tracking vegetation water stress and dryness of soils and MODIS supplies LST data vital for heat stress detection. The system acquires data from Google Earth Engine (GEE) through its satellite data extraction capabilities and satellite data preprocessing functions alongside Google Colab as the deep learning platform for training and evaluation procedures. The STGNN model unites satellite perception with geographic pattern knowledge in a chronological order to forecast drought levels and soil moisture change across time while we trained a Long Short-Term Memory (LSTM) deep learning algorithm to analyze time sequences. The STGNN model evaluates complicated land patterns thus produces superior generalization capability and understanding relative to simple prediction approaches.

Keywords—Harmonized Landsat Sentinel (HLS), STGNN, LSTM, Google Earth Engine, MODIS

INTRODUCTION

Drought represents a recurring and catastrophic natural disaster which endangers world food markets while putting significant pressure on water systems and jeopardizing ecosystem well-being. The lack of abnormally low rainfall during long-duration periods defines drought and produces significant socio-economic as well as environmental effects that affect agriculture and industry and the survival of people. The implementation of effective drought adaptation and mitigation approaches depends on proper and prompt drought dynamics observation. The measurement of drought

conditions represents an urgent need for India because agriculture depends completely on monsoon rainfall while many people live through rainfall-based farming. The diverse climatic regions of India are frequently subjected to drought which affects the country as one of the most susceptible nations to natural hazards. Drought-related challenges in India result in many serious effects that include killed crops and dead animals combined with inadequate water supply and societal tension.

Within India, the state of Maharashtra, and specifically the Marathwada region, stands out as a particularly drought-prone area. Marathwada experiences frequent droughts because of its dry semi-arid conditions and unpredictable rainfall variations which resulted in three significant drought periods: 1972, 2000, 2003, and a long-lasting drought between 2012 through 2016. The agrarian economy together with the rural residents' livelihoods endured major challenges throughout these dry periods. Rain-fed agriculture in the region suffers from high vulnerability to monsoon uncertainties because the area depends on rain for agriculture cultivation.

Traditional methods of drought evaluation which use on-ground observations show limited spatial extent and require extensive manual work which restricts prompt intended actions. The ability of satellite remote sensing to produce continuous observations over vast areas has established it as a fundamental drought monitoring technology during the past few years. Retrieval of valid information from multi-sensor satellite data demands the use of advanced analytical methods and technology.

The research utilizes combined HLS and Sentinel-1 and multi-time frame capability from the Sentinel constellation along with soil moisture reading and land surface information from Sentinel-1 Synthetic Aperture Radar data to analyze the Marathwada region in India. The integration of SAR data within a deep learning pipeline and implementation of STGNN enables improved prediction of drought while filling the shortcomings of data from previous research.

STGNNs demonstrate strong potential for enhanced drought analysis by processing complex environmental dependencies and therefore, improving both identification and prediction and also, reduces the vulnerability of this important ecological region. The integrated approach will be analyzed

through this paper to demonstrate its success in improving drought events, understanding and managing drought in Marathwada's critical agricultural region of India.

LSTMs specialize in sequential data modeling because they derive data from specialized recurrent neural networks (RNNs), these models learn long-term temporal dependencies of the data. LSTMs also show suitable performance for drought prediction because they effectively track multiple-timeframe sequences between climatic and remote sensing datasets that include vegetation indices and soil moisture levels and precipitation patterns. The paper develops a deep exploration of Long Short-Term Memory models that use combined data from HLS and Sentinel-1 to model Marathwada area conditions. The deep learning architecture generates practical remarks to establish warning systems, resource plans and agricultural adaptation strategies to build better drought resistance and sustainable land management within the region.

LITERATURE REVIEW

Deep Learning-Based Harmonization and Super-Resolution of Landsat-8 and Sentinel-2 Images (2024) [1]. Derived from this paper development a deep learning-based drought monitoring framework integrating multiple remote sensing datasets. The study employs CNNs to model drought indices. This study presents a deep learning-based pipeline to harmonize spectral and spatial differences between Landsat-8 and Sentinel-2 (Harmonized Landsat and Sentinel) satellite imagery. By enhancing Landsat-8's spatial resolution and integrating data from both satellites, the pipeline increases cloud-free image availability by 21% annually. It also improves NDVI correlation between the two satellites by 4.9%, leading to better performance in crop segmentation tasks. The 100M parameter model, trained on European data, generalizes well globally with minor limitations [1].

Construction of an Integrated Drought Monitoring Model Based on Deep Learning Algorithms (2023). This study explores deep learning for drought monitoring using multi-source remote sensing data, including MODIS and CHIRPS, to analyze precipitation, vegetation, temperature, and soil moisture. The model effectively captured drought spatial patterns, offering a robust approach for regional drought monitoring [2].

Drought Risk Assessment Using Machine Learning and Remote Sensing Data (2023). It surveys various drought prediction models, from traditional statistical methods to advanced machine learning and deep learning techniques, highlighting their superior accuracy. The study explores the integration of remote sensing data, meteorological observations, hydrological modeling, and climate indices to enhance prediction reliability [3].

Utilization of multi-sensor data to analyze drought trends at a local scale. It integrates MODIS and Sentinel-2 images to derive vegetation health indices. Also, demonstrates the importance of multi-sensor fusion for drought monitoring and provides methodologies for using vegetation indices in drought assessment [4].

Exploring the potential of Sentinel-1 SAR for detecting drought-induced forest damage; Highlights Sentinel-1 SAR's role in monitoring vegetation water content and provides methodologies for analysing SAR backscatter signals in drought conditions [6][7].

Fusion of Sentinel-1 SAR and Sentinel-2 MSI Data for Accurate Urban Land Use-Land Cover Classification in Gondar City, Ethiopia [10]. Demonstrates data fusion techniques for Sentinel-1 SAR and Sentinel-2 MSI, which aligns with your research on harmonized Landsat-Sentinel datasets using machine learning classifiers (SVM, RF), providing insights into algorithm performance for multi-source remote sensing data and highlights accuracy improvements when combining optical (MSI) and SAR data, which is crucial for enhancing drought prediction models. Also suggests implementation of deep learning-based models like STGNNs and GANs for improved accuracy in spatiotemporal drought assessment.

All the papers and studies have suggested that lot of advancements can be done. Time-series analysis using deep learning (e.g., STGNNs, LSTMs) is needed for real-time drought monitoring and SAR-based soil moisture retrieval (from Sentinel-1) should be integrated with deep learning models for better drought prediction. Advanced deep learning models (STGNNs) remain largely unexplored for spatiotemporal drought prediction and that the integration of CNNs, transformers, and recurrent models could improve drought classification accuracy. Combining optical (Sentinel-2 MSI, Landsat) and SAR (Sentinel-1) has enhanced classification accuracy, as seen in urban land use studies.

A detailed review of all these above papers and research studies has brought us to following conclusions, addressing following research gaps:

The fusion of harmonized Landsat-Sentinel data with Sentinel-1 SAR provides a strong foundation for improving drought monitoring.

Traditional machine learning models (RF, SVM) have been effective but lack the predictive power of deep learning architectures. Spatiotemporal drought prediction using deep learning models remains underdeveloped in the current research landscape.

The following limitations are to be targeted:

- Developing deep learning models (STGNNs) to predict drought conditions over time
- Enhancing multi-source data fusion by integrating harmonized Landsat-Sentinel with Sentinel-1 SAR soil moisture estimates.
- Implementing real-time monitoring systems using Google Earth Engine (GEE) and deep learning frameworks.
- Combining remote sensing with meteorological and hydrological data for improved drought severity classification.

METHODOLOGY

A. Study Area

Marathwada, a region within the Indian state of Maharashtra, is notably susceptible to drought. Its dry, semi-arid environment and erratic rainfall contribute to frequent dry spells, including significant drought periods in 1972, 2000, 2003, and a prolonged one from 2012 to 2016. These conditions have severely impacted the agrarian economy and the well-being of rural inhabitants. The region's dependence

on rain-fed agriculture makes it particularly vulnerable to the uncertainties of the monsoon.

This research integrates data from multiple sources and advanced techniques. It leverages the combined capabilities of HLS and Sentinel-1 imagery, utilizing the Sentinel constellation's multi-temporal data. Additionally, it incorporates soil moisture readings and land surface information derived from Sentinel-1's Synthetic Aperture Radar (SAR) data. By integrating this SAR data within a deep learning pipeline and employing a Spatio-Temporal Graph Neural Network (ST-GNN) and LSTM, the study aims to achieve more accurate drought prediction, overcoming limitations found in previous research.

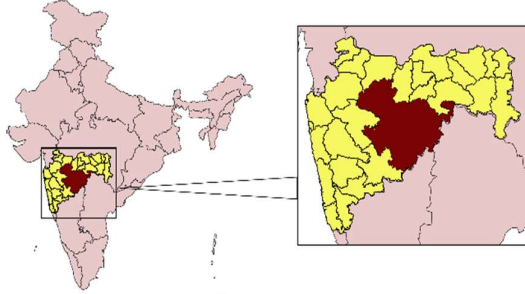


Fig 1. Study Area

B. Dataset and preprocessing

The proposed methodology is described in the figure 2. The following steps are implemented to process the satellite images:

The dataset for drought prediction was collected from multiple satellite sources, including Sentinel-1, Harmonized Landsat and Sentinel-2 (HLS), and MODIS. Google Earth Engine (GEE) was utilized for data acquisition, and Google Colab was used for processing and model training. The details of the datasets are as follows:

1. Sentinel-1 (SAR Data): Sentinel-1 is a radar imaging satellite service controlled by The European Space Agency through its Copernicus program. Synthetic Aperture Radar (SAR) is used by Sentinel-1 to deliver detailed all-weather and day-and-night Earth surface observations. The satellite provides valuable data to track both floods and land deformation and vegetation health in addition to soil moisture.

- SAR technology operates as an active remote sensing method which utilizes microwave signals to generate detailed surface images through Earth observation. SAR technology has the advantage of producing images through clouds and can provide analysis under all weather conditions. Therefore, it proves beneficial for applications such as drought assessment and forest evaluation and emergency response operations.
- Bands VH and VV were used to compute radar-based vegetation indices.
- In VH-polarization (Vertical-Horizontal), the polarization enables analysis of vegetation structure because it detects scattering from leaves and branches within vegetative areas.

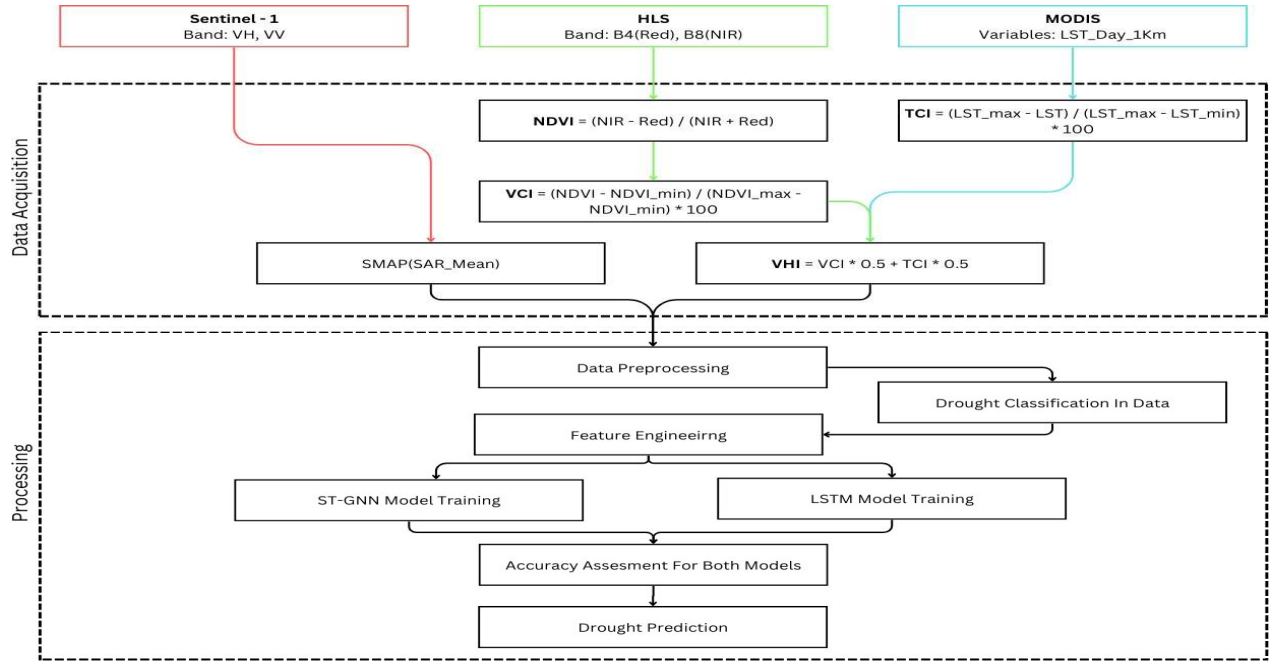
- In VV Polarization (Vertical-Vertical), the radar both sends and receives signals with vertical polarization. The sensor detects both surface roughness and water bodies with high sensitivity because of its ability to monitor soil moisture and water-related modifications.

2. HLS (Optical Data): HLS is a dataset that combines image data from NASA's Landsat and ESA's Sentinel-2 satellites.

- The satellite data offers standardized surface reflectance information at high resolution levels.
- The surface reflectance data measured by visible spectrum and near-infrared and shortwave infrared wavelength bands form the basis of the Harmonized Landsat and Sentinel-2 (HLS) optical data. Analysis of vegetation health together with land cover changes and water content is supported through the use of these bands.
- NDVI was calculated through the usage of bands B4 (Red) and B8 (NIR). The red hue area within the electromagnetic spectrum matches with Band B4. NDVI uses this band because it responds to vegetation consumption of sunlight throughout its operational spectrum. Vegetation absorbs red light from its healthy state yet shows more reflection when it experiences stress or extent plant reduction.
- Near-infrared (NIR) reflectance that Band B8 detects shows strong reflection from healthful vegetation. Vegetation analysis heavily depends on NIR band for its analysis since plants reflecting more NIR light possess high chlorophyll content. The index tool applies this wavelength to calculate NDVI which monitors vegetation health together with moisture conditions.

3. MODIS (Thermal Data): MODIS sensor operated by the NASA, as a primary instrument on Terra and Aqua satellites which delivers Earth observations using various spectral bands across the globe.

- The satellite provides data related to vegetation identification as well as observations of land cover and sea surface temperature and atmospheric information.
- Land Surface Temperature (LST_Day_1Km) was extracted for temperature-based drought assessment of the data.
- The LST measurement obtains surface temperature readings by infrared sensors located on Earth. Surface heat stress detection and drought severity evaluation and evapotranspiration determination rely on the 1 km resolution LST data provided by MODIS. Surface heat stress increases with higher LST values and droughts might develop under such heat stress conditions.



C. Computation of Drought Indices:

Further the dataset was trained and used to compute various indices:

1. The process of estimating soil moisture through Sentinel-1 SAR data requires analysis of backscatter coefficients (σ°) in VV and VH polarizations. C-band SAR data supplied by Sentinel-1 is well known for its potential in soil moisture retrieval because it detects changes in surface roughness and dielectric characteristics.

2. NDVI serves as a common tool for assessing vegetation health because it measures the distinction between Near-Infrared (NIR) and Red wavelengths reflection.

$$NDVI = \frac{NIR - R}{NIR + RED}$$

Where:

NIR: Near-Infrared reflectance (HLS B8)
Red: Red reflectance (HLS B4)

3. The VCI (Vegetation Condition Index) normalization process uses present NDVI data along with historic NDVI reference to distinguish drought conditions.

$$VCI = \frac{NDVI - NDVI_{min}}{NDVI_{max} - NDVI} \times 100$$

Where:

$NDVI_{min}$: The lowest observed NDVI value over a given time period.

$NDVI_{max}$: The highest observed NDVI value over the same time period.

4. The evaluation of drought through land surface temperature

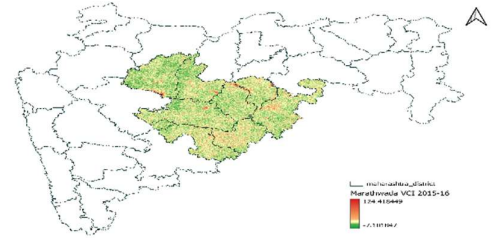


Fig 3. VCI during 2015-16

anomalies happens using TCI (Temperature Condition Index). Higher temperatures serve as an indicator that vegetation faces stress.

$$TCI = \frac{LST_{max} - LST}{LST_{max} - LST_{mi}} \times 100$$

Where:

LST: Land Surface Temperature from MODIS

LST_{max} and LST_{min} : Maximum and minimum LST values over time.

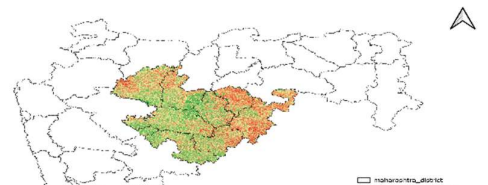


Figure 4. VCI during 2020-21

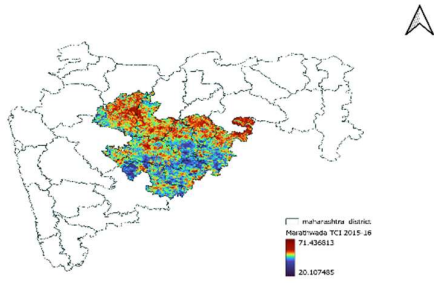


Fig 5. TCI during 2015-2016

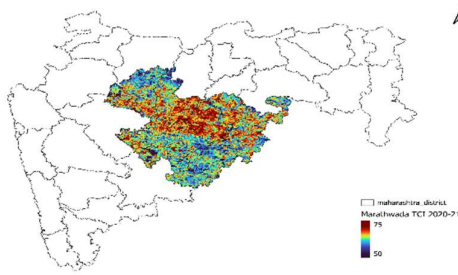


Fig 6. TCI during 2020-2021

5. The combination of VCI (Vegetation Condition Index) and TCI forms the VHI (Vegetation Health Index) for total drought evaluation.

$$VHI = 0.5 \times VCI + 0.5 \times TCI$$

Where: a lower VHI value signifies extreme drought

D. Models Training and Evaluation:

i) ST-GNN overview

The Spatio-Temporal Graph Neural Network functions as a deep learning solution which extracts spatial and temporal signals from structured graph data (Graph Convolutional Network). The model demonstrates effectiveness when analyzing connected data that evolves periodically such as the evaluation of soil moisture as well as drought forecasting. The model concludes geographic features of neighbouring influences with time-based patterns to forecast future soil moisture levels.

To train the model the dataset is converted into a graph structure, where:

- Nodes represent different geographical locations (e.g., pixels in a satellite image).
- Edges connect nearby locations based on distance or similarity (e.g., adjacency matrix).
- Node Features include soil moisture, Sentinel-1 SAR backscatter (VV, VH), NDVI, LST, precipitation, etc.

An adjacency matrix function as a network representation tool for recording location-to-location connections. The model obtains insights about moisture operation across connections because of this representation method. Node features from the Graph Convolutional Network (GCN)

receive processing through neighbouring information. The model determines new moisture measurements through comparing moisture levels between neighbouring nodes. Wet soil appears through high NDVI combined with low backscatter while dry soil shows itself as low NDVI and high backscatter.

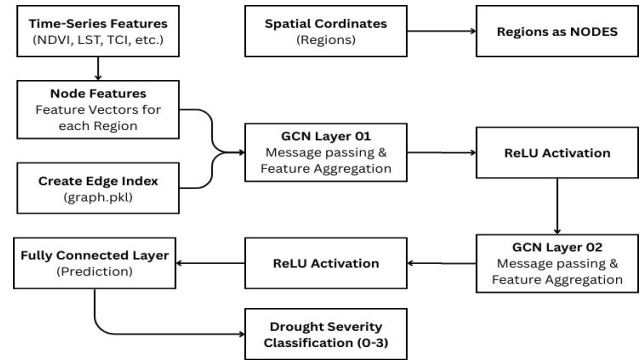


Fig 7. Working of STGNN

The GCN distributes soil moisture information between neighboring nodes to identify wettest areas by high values and driest regions by low values based on satellite remote sensing and geographic relationships.

ii) LSTM overview

Used LSTM stands as a distinguished form of Recurrent Neural Network which utilizes sequential data to track land surface temperature through its Long Short-Term Memory structure that handles time-related patterns. The key function of LSTM is to model temporal dependencies which explain soil moisture changes over time through a combination of past weather patterns and vegetation along with SAR backscatter measurements.

LSTM Through LSTM the model acquires an ability to analyze temporal soil moisture patterns by studying historic patterns. ST-GNN makes accurate soil moisture predictions through its combination of Long Short-term Memory (LSTM) with spatial modeling (GCN) thus providing valuable solutions to drought monitoring along with agricultural planning.

The data visualizes through heat maps on the geographical map using the model where dark blue indicates water-logged zones while red indicates dry areas.

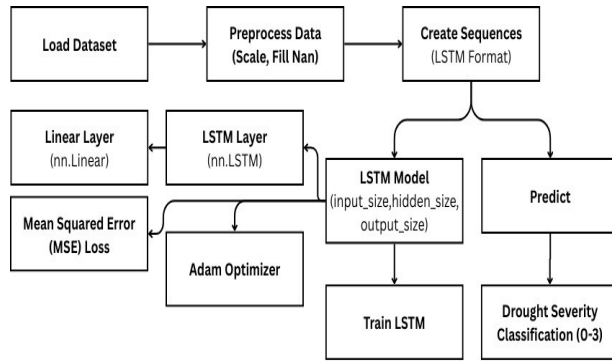


Fig 8. Working of LSTM

iii) Evaluation of Drought

The drought value computes information stemming from HLS (utilizing NDVI and EVI indicators) and Sentinel-1 (backscatter of VH/VV along with soil moisture measurements) to unify measurements of surface characteristics and vegetation health. The deep learning models LSTM or STGNN analyze processed inputs to establish connections between chronological patterns common in drought occurrences. The prediction model produces drought severity measurements or classifications (no drought or moderate or severe) through evaluation of standardized indices like SPI, SPEI or custom metrics. The system allows dynamic operational drought tracking and forecasting specifically meant for the Marathwada area while additionally strengthening early alert systems and developing agriculture strategies that are resilient to climate changes.

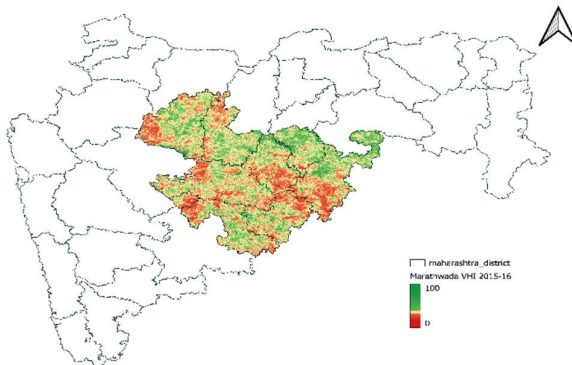


Fig 9. Drought Depiction during 2015-2016

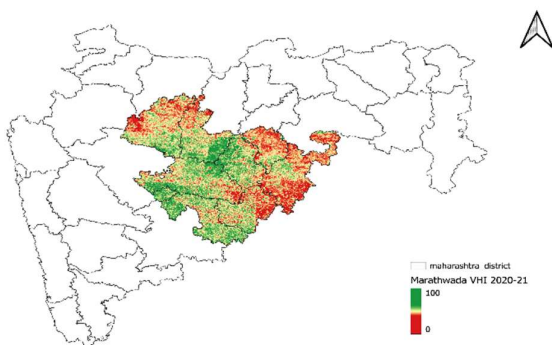


Fig 10. Drought Depiction during 2020-2021

CONCLUSION

The proposed research investigates deep learning prediction of drought phenomena using Synthetic Aperture Radar analyses combined from Harmonized Landsat-Sentinel and Sentinel-1 satellite data. Our method effectively captured the space-time changes in soil moisture data through its combination of optical and radar remote sensing information. Implementing the Spatial-Temporal Graph Neural Network (STGNN) model has developed an ability to detect complex patterns which connected the spatial and temporal domains resulting in more accurate and precise drought predictions.

The Google Earth Engine tool served as the foundation platform for implementing both data acquisition and preprocessing operations through its efficient handling of satellite image datasets. The deep learning workflows benefited from Google Colab because it provided scalable and flexible experimental capabilities to develop the models. The experimental data shows that STGNN surpasses standard models through its ability to use spatial temporal relationships between geographic zones and time-dependent evolutions.

The research adds value to climate resilience science through its data-based automated solution for predicting early signs of drought. The developed system can provide assistance for agricultural planning as well as water resource management and disaster mitigation approaches.

The method we presented creates an approach that uses advanced deep learning methods with abundant satellite data to establish scalable and intelligent drought prediction systems.

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